

Generational differences in internal migration

Matteo del Piano ^{*} [†]

Last update: 07/2026

Abstract

This paper studies differences in internal mobility across migrant generations using data from the CPS ASEC. I estimate a linear probability model and find that first-generation migrants are on average more mobile across states while second-generation migrants are indistinguishable from natives. Second-generation migrants are less mobile in short- to medium-distance movements. This has important labor market implications because of migrants' role in wages and welfare in local labor markets and their contribution to smoothing local demand shocks. These average patterns mask substantial heterogeneity both in terms of workers' ethnicity and skill level; within these two dimensions, estimates provide evidence for different mobility gradients. With the addition of a gravity model, I document that dyadic geographical characteristics, destination-specific features, and the ethnic composition of the corridor are significant predictors of migration flows within each origin-destination pair.

1 Introduction

Internal migration is an important mechanism of spatial adjustment within the boundaries of a country. Compared to international migration, internal spatial sorting is more related to workers' expectations on plausible labor market outcomes, mainly wages and employment; on the cost side, the reallocation imposes the trade-off of remaining settled in the community in which workers have been residing since arrival.

This paper targets one single empirical question: are there differences in internal mobility of first-generation migrants compared to second-generation migrants and natives? Following from this principal research motivation: are these differences constant across ethnicity, skill level, and geographical definitions?

^{*}Free University of Bolzano, Department of Economics and Management

[†]University of Trento, Department of Economics and Management

The importance of this paper is related to multiple factors.

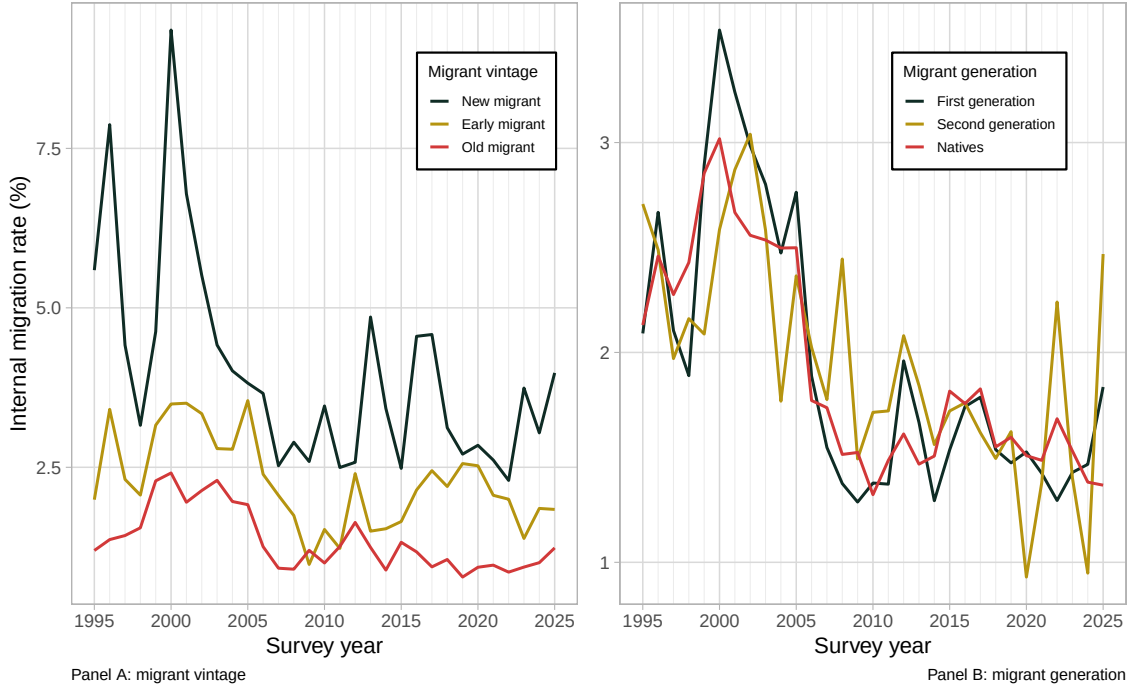
Differences in mobility are crucial to understanding how migration, both in the international and internal sense, shapes and impacts labor markets, as several studies in this literature have emphasized. Higher response rates are fundamental in mitigating the negative consequences of a labor demand shock (Cadena and Kovak 2016); internal migration explains wage convergence across states in the US (Amior 2025; Borjas 2001); and it significantly impacts salaries and local economic conditions (Aubry et al. 2026).

These previous papers, however, limit the comparison to analyzing each outcome and variable at the natives vs. migrants level. The extension toward the second generation is interesting as this group is big enough to have a meaningful impact through migration on labor markets, but it is different in composition with respect to the other two. Hence understanding which pattern it points towards, first-generation or natives (convergence hypothesis, i.e., whether 2nd gen behavior converges to that of natives or remains closer to 1st gen), is relevant both to distinguish potential unobserved heterogeneity from previous studies as well as for policy implications when designing interventions that tackle a specific source of migration. To get a better sense of the mobility of workers, expanding the analysis toward second-generation migrants is important when trying to infer from which source the migration channel will impact the economy.

This paper is the first one, to my knowledge, that augments the analysis of these differences by considering second-generation migrants related to first-generation migrants and natives. The closest work to this one has been produced by Braun and Görts 2025, in which the authors have studied the impact of the European debt crisis on individuals' locations across European countries, while expanding the mobility response to labor demand shocks towards the 1.5 migrant generation (i.e. respondents who settled in the foreign country while they were young).

Using data from the CPS ASEC (Annual Social and Economic Supplement), I estimate a linear probability model and a gravity model. The former describes empirical differences in internal mobility across different geographic definitions, state moves, across- and infra-county moves, and decomposes the average mobility by studying the ethnic and skill gradient. The latter identifies the most relevant economic channels that act as principal pull drivers across several dimensions: dyadic characteristics, destination pull factors, and corridors' ethnic similarity. Workers in the sample are categorized into migrant-generation groups according to the following rule: foreign-born individuals who have entered the US are labeled as first-generation migrants. Workers who are born in the US but have at least one foreign-born parent are second-generation migrants, and US-born individuals with both US-born parents are considered natives.

Figure 1: Internal migration rate.



Note: Internal migration rate is calculated as the share of individuals that report a different state of residence today compared to the previous year. The left panel shows the decomposition of foreign-born migrants according to arrival categorization in the US; the right panel instead shows the internal migration rate conditional on the three generation categories.

Figure 1 displays interstate migration rates conditional on two dimensions.

Panel A on the left shows internal migration rates, computed as the share of individuals who have changed their residence state in a specific year, for new, early, and old first-generation migrants. A migrant is considered new if he or she entered the US no more than 5 years preceding the survey; an early migrant is a foreign-born worker who has been in the country for 5 to 15 years; old migrants are those who have been in the country for more than 15 years. Panel B on the right shows the same internal migration rate conditional on the three generations.

New migrants are the most mobile group in the sample. This is consistent with a large fraction of studies that have analyzed internal migration in the US. Early and old migrants follow directly, suggesting that there is a monotone relationship, in aggregate terms, between time spent in the US and internal migration. Looking at internal mobility rates, the picture is a bit more puzzling. Looking just at the descriptive series, it is unclear whether one group dominates the other.

This paper provides empirical evidence that describes higher interstate migration rates for first-generation migrants with respect to the other two groups. Second-generation migrants do not display interstate mobility patterns different from those of natives. When looking at shorter distance moves, I find that first- and second-

generation migrants are less mobile in inter-county migration within the same state, compared to natives, and second-generation migrants are less mobile in within-county moves. These results are presented in Section 4 of the paper.

In the second part of the paper I complement the first piece of evidence with estimates taken from a gravity model. With that specification I am able to find which economic variables are the most relevant when predicting the bilateral origin-destination flows in terms of dyadic characteristics, destination pull factors and corridor ethnic similarity.

This paper aligns with the literature on internal migration in the US. The empirical evidence that I uncover is consistent with previous studies with respect to the main evidence of the paper. The addition of the second generation, the heterogeneity decomposition, and the general framing proposed in this study, however, bring a new set of descriptive evidence on this phenomenon that adds value to previous findings.

The rest of the paper is organized as follows. Section 2 provides a literature review and the motivation for this work. Section 3 provides a description of the sample, including average workers' characteristics and their mobility. Section 4 presents the results from the linear probability model estimates and the decomposition of the unobserved heterogeneity in migration likelihood. In this context some robustness estimates are presented. Section 5 describes the gravity model framework and presents the estimates from different specifications of the model. Section 6 concludes.

2 Literature review

The starting point for this paper is the seminal contribution by Sjaastad 1962. In that article, Sjaastad framed for the first time the migration decision as an individual maximization problem: workers move when the expected net returns at the destination exceed those at the origin, after accounting for migration costs. The costs that each potential migrant needs to consider are several, both in terms of monetary and non-monetary dimensions. The former are easier to quantify; they include travel costs, acquiring information, and settling in the destination. The latter are more endogenous and individual-specific, for example the cost of leaving the local community in which an individual grew up and the shadow cost of moving, which includes the loss of the network and the labor market value of the individual's work.

Natives and migrants are two different groups, and they behave differently when comparing internal migration outcomes. On the one hand, several studies have emphasized the higher mobility of foreign-born workers who have entered the US (Basso and Peri 2020; Borjas 2001; Cadena and Kovak 2016; Maré, Morten, and Stillman 2007), and others have provided similar results when looking at this dimension in

European countries (Schündeln 2014).

This differential in migration elasticities when considering both the response to positive or negative shocks in the labor market has significant consequences. For example, in Cadena and Kovak 2016, the impact of the subprime crisis of 2008 in the US induced a higher reallocation of workers, as the economic consequences of the crisis have been reflected in employment probabilities rather than wages. The higher Mexican mobility significantly reduced job losses for natives, and thus mitigated the negative impact on that specific group. Differences in migration elasticities are crucial also regarding wage convergence and adjustment, as motivated in Amior 2025.

All this previous literature does not include differences along the generational dimension in the analysis, which leaves us, at most, with an intuition on how these arguments translate when considering that specific type of worker.

These differences in elasticities are a composition of different channels.

The first thing is that migrants have access to a network of individuals who have already experienced settlement in that location. The relevance of ethnic enclaves has been demonstrated in several studies to be one of the most important predictors (Munshi 2003; McKenzie and Rapoport 2010). The presence of migrants who have already settled in a new state helps attenuate the cost of migration, both by reducing the costs of acquiring information and by providing access to an established network of workers who have settled in a community. While networks reduce the cost of the first move, they may also act as an anchor that discourages subsequent moves once a worker is embedded in the enclave.

Another argument about the difference is that first-generation migrants are a self-selected group of individuals who are less averse to risk. This implies that their behavioral attitude may reveal a higher internal mobility, *ceteris paribus*, when compared with natives.

Second-generation migrants are somewhat in the middle of this generational difference between foreign-born workers and natives. On the one hand, being born in the US and having grown up in their communities, they should have developed a location-specific attachment, which should reflect a higher cost of leaving their state. On the other hand, the endogenous culture in which they grow up may place a higher importance on mobility. If the local attachment dominates, then we can expect to find mobility patterns in line with those of the native group, and verify the convergence hypothesis; but if the network effect and the inter-generational attitude towards migration are transferred from the parents to the US-born individuals, then we should expect a behavior pointing towards the first generation.

The added value to this paper, then, is the description of mobility patterns specifically including the second generation in the analysis. The only similar study that has

tried to expand this dimension is Braun and Görts 2025, in which the authors found that mobility responses after the European debt crisis for the 1.5 generation were in between those of foreign-born workers and natives. This paper extends the analysis to the second generation proper — US-born individuals with at least one foreign-born parent — distinct from the 1.5 generation, and focuses on internal mobility within the US.

3 Data

3.1 Source

The main source of data is the IPUMS CPS Annual Social and Economic Supplement (ASEC) data files ¹. The sample covers the time span between 1995 and 2025, and the unit of analysis is full-time workers aged between 25 and 54, who are not enrolled in high school or college and do not reside in close quarters. This filtering is crucial to correctly identify workers in their prime age and to focus on labor-driven mobility.

I have used CPS microdata files for a specific reason: compared to the standard census ACS source, they contain specific information about respondents' parental background. Having direct access to parents' characteristics allows me to categorize each worker in the sample into one of the three migrant generations without using probability models or imputation from declared ancestry or ethnicity.

The principal variable of interest is the migrant generation variable. The levels created identify respondents across three categories according to these conditions: respondents who were born outside US boundaries are coded as first-generation migrants, individuals who have at least one foreign-born parent and are US-born are defined as second-generation migrants, and lastly workers who are US-born with both parents US-born are coded as natives. Thus, individuals with one US-born and one foreign-born parent are classified as second generation, following the standard Census/Pew definition. Unfortunately, there is not enough data on grandparents to continue the coding chain.

Additional data have been used in the paper. The other source is census ACS data from the year 1990 to the present. These data have been combined extensively in the second part of the paper to build state-specific characteristics used as pull factors in the gravity model.

¹*Sarah Flood, Miriam King, Renae Rodgers, Steven Ruggles, J. Robert Warren, Daniel Backman, Etienne Breton, Grace Cooper, Julia A. Rivera Drew, Stephanie Richards, David Van Riper. Integrated Public Use Microdata Series, Current Population Survey: Version 13.0 [dataset]. Minneapolis, MN: IPUMS, 2025. <https://doi.org/10.18128/D030.V13.0>*

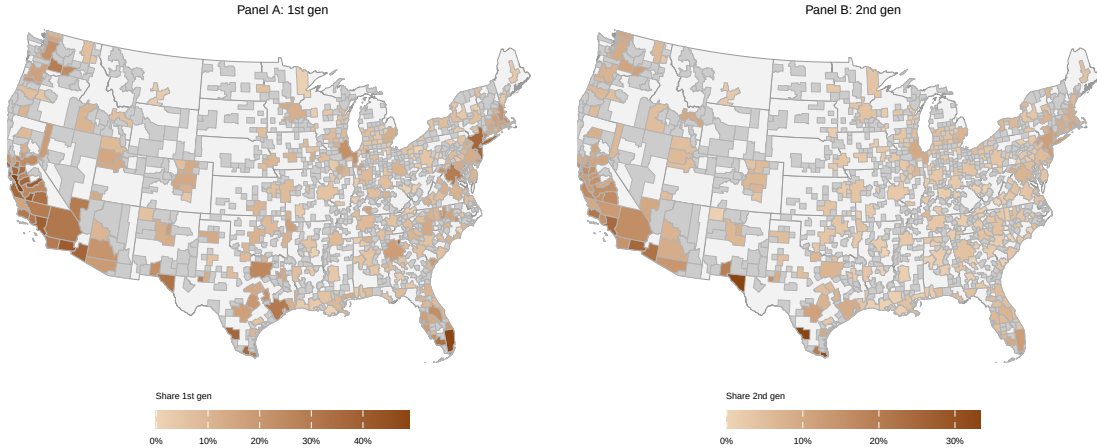
3.2 Descriptive statistics

Migrants' distribution in the US is clustered in specific metropolitan statistical areas. Figure 2 maps, in two panels, the share of first-generation migrants and second-generation migrants over the native population in US metropolitan statistical areas (i.e. aggregated urban areas with more than 100,000 residents).

The spatial distribution of migrants is consistent with previous findings in the literature. Foreign-born individuals tend to cluster into specific metropolitan areas in which previous migrants have already settled, thus leveraging the network effect. This fact is not surprising given that many papers have already found this agglomeration effect, and it should be interpreted as an attenuation of the movement costs involved in migration (Munshi 2003; McKenzie and Rapoport 2010).

The distribution of the second generation displays an interesting pattern. It seems that second-generation migrants are clustered in the states where the previous generation has settled. As half of the second generation is born to two foreign-born parents (51%), this seems a reasonable pattern that emerges from the data. Additionally, appendix figure 10 shows that new arrivals are concentrated in some port cities, in which the presence of already settled immigrants is consolidated.

Figure 2: Migrant concentration in Metropolitan Statistical Areas.

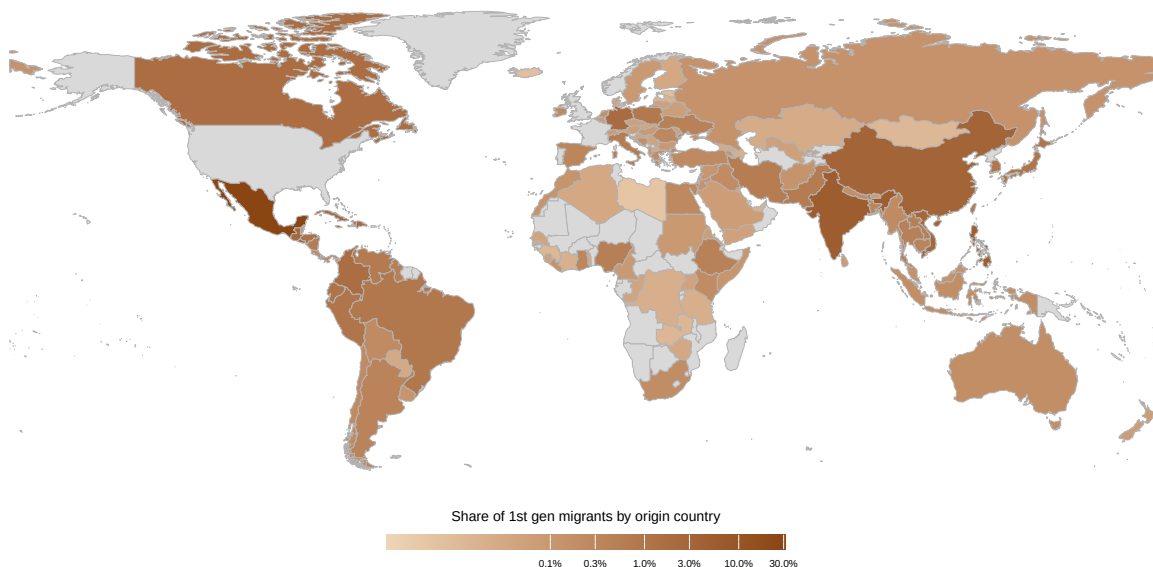


Note: Sample restricted to respondents with a valid metropolitan-area code, aggregated over all survey years (1995–2025) and weighted by ASEC sample weights. Metropolitan areas follow the 2013 OMB Core-Based Statistical Area delineations, obtained via a crosswalk from the CPS metropolitan-area code. Panel A reports the weighted share of first-generation migrants over the total MSA population; Panel B the analogous share for second-generation migrants.

The most popular birthplace for first-generation migrants is Mexico (figure 3). This is not surprising as the foreign-born population in the US is dominated by the share of Mexicans over the total population. It is interesting to notice, however, the

large share of individuals who come from China and, more generally, from the Asian continent. This fact is important, as the Asian cohort will prove to be among the most mobile ethnic groups in interstate migration (see Section 4.2; Postel 2026).

Figure 3: Distribution of first-generation migrants by country of origin.



Note: Sample restricted to first-generation (foreign-born) respondents. Country of birth is matched to ISO3 codes; historical or sub-national codes no longer in use (e.g., Czechoslovakia, Yugoslavia, England/Scotland/Wales/Northern Ireland, the Azores) are manually mapped to their present-day country. Shares are computed over the total first-generation sample; the color scale uses a $\log_{10}$ transformation to accommodate the highly skewed distribution of origin countries.

The remaining portion comes principally from other Latin American countries and Canada.

The sample is quite balanced across the three generations in terms of demographic characteristics (table 1).

The most represented group is composed of natives, who make up 75.7% of the sample. The first generation is the second highest, with 17.6%, and the smallest represented group is the second generation, with 6.68%.

The majority of differences arise in economic characteristics, ethnic composition and co-residence status. Foreign-born individuals are less skilled and earn lower labor market wages compared with the other two groups; they also hold the highest share of co-residing status with their partner. This last fact is interesting because it points to the possibility that savings and income are sent back as remittances; additionally, it will be necessary to consider this fact when estimating the linear probability model, as singles are in general more mobile than married couples (see Section 4.3).

Table 1: Sample description.

Characteristic	First gen (born abroad) N = 272,296	Second gen (at least 1 US-born parent) N = 103,336	native N = 1,170,159
Age (Years)	40 (8)	38 (8)	40 (8)
Years of education	12.75 (4.13)	14.40 (2.57)	14.10 (2.37)
Total personal income (2010 USD)	49,228 (61,332)	60,575 (64,961)	57,436 (61,281)
Sex			
male	(61%) 165,400	(56%) 57,377	(55%) 648,866
female	(39%) 106,896	(44%) 45,959	(45%) 521,293
Marital status			
married, spouse present	(66%) 179,699	(56%) 57,743	(65%) 756,700
married, spouse absent	(3.8%) 10,381	(1.3%) 1,304	(0.9%) 10,471
separated	(3.1%) 8,547	(2.2%) 2,260	(2.1%) 25,072
divorced	(7.0%) 18,995	(9.0%) 9,251	(12%) 136,812
widowed	(0.8%) 2,244	(0.7%) 681	(0.9%) 9,949
never married/single	(19%) 52,430	(31%) 32,097	(20%) 231,155
Ethnicity			
White non-Hispanic	(17%) 46,873	(41%) 41,586	(81%) 924,670
Black non-Hispanic	(6.9%) 18,695	(3.8%) 3,890	(11%) 128,719
Asian non-Hispanic	(25%) 67,096	(15%) 15,419	(1.3%) 15,304
Mexican Hispanic	(31%) 83,934	(28%) 29,158	(3.7%) 42,837
Hispanic non-Mexican	(20%) 54,615	(12%) 12,446	(2.8%) 31,876
Unknown	1,083	837	26,753

¹ Mean (SD); (%) n

Note: Continuous variables (age, years of education, total personal income) are reported as mean (standard deviation); categorical variables (sex, marital status, ethnicity) as column percentage (count). Total personal income is deflated to 2010 USD using the CPI-U.

Lastly, table 2 displays the unconditional mean for the dependent variables that will be used in the linear probability model.

Table 2: Average mobility for each group.

Characteristic	First gen (born abroad) N = 272,296	Second gen (at least 1 US-born parent) N = 103,336	native N = 1,170,159
State move	1.8%, 4,811	1.9%, 1,949	1.9%, 21,937
State move 5 year	7.0%, 2,881	7.9%, 1,336	8.1%, 14,227
Division move 5 year	4.9%, 1,741	5.7%, 817	5.0%, 6,994
County movement	8.3%, 22,484	7.7%, 7,996	7.4%, 86,928
Across counties movement	1.9%, 5,074	2.1%, 2,166	2.4%, 28,526

¹ %, n

Note: Unconditional share of respondents (% , count) reporting Yes for each mobility outcome, by migrant generation. State move and the infra-county/infra-state indicators are measured over a 1-year horizon; state move (5yr) and division move (5yr) are restricted to the 1995, 2005, 2015, 2020 and 2025 CPS-ASEC waves, the only years for which 5-year migration history is collected.

The first line displays a similar unconditional mean across all groups. In the results from the empirical model I will show that, after taking into account the contribution of state and year fixed effects and the other controls, first-generation migrants display a higher interstate mobility. A second note must be added to the state mobility at the 5-year level. Those shares mask a bias because there is a large share of foreign-born individuals who report being abroad 5 years prior².

²29.5% of new migrants reported living abroad 5 years prior to the survey.

3.3 Heterogeneity in mobility rates

Interstate mobility rates display an important heterogeneity across different generations on both ethnicity and skill dimensions.

In table 3 it is possible to visualize the internal migration rate measured in different outcomes. The first generation is further decomposed into new, early, and old migrants (respectively, less than 5 years in the US, between 5 and 15 years, and more than 15 years).

New migrants and second-generation migrants display U-shaped rates relative to their skill level when considering short-term mobility: less educated individuals are mobile because of a higher job rotation on average, while more educated workers are mobile because their labor market opportunities are concentrated. Long-term moves lose the U-shaped pattern relevant for the short term; they are, in general, monotonically increasing in human capital.

Table 4 replicates the structure relative to workers' skill, but this time decomposing the rates conditional on the individuals' reported ethnicity.

The most important feature of this table is the level of the interstate migration rate for the pool of Mexican migrants, which displays the lowest rate among all ethnic groups. This fact, in addition to the relative importance of their share among foreign-born individuals, explains the low unconditional mobility found before. Looking at the data, it is also possible to visualize the important contribution that Asian workers make to internal migration, as they are the most mobile group among new migrants and early migrants. On average, the dispersion of the rate does not display any particular pattern in this case.

These descriptive patterns motivate the formal heterogeneity analysis in Section 4.2, where I estimate generation-specific gradients along skill and ethnicity conditional on demographic and geographic controls.

4 Empirical model

This section estimates conditional differences in mobility across generations using a linear probability model.

I have created 3 different measures for the dependent variable used in this model. State move is a proxy for short-term mobility and equals one when a worker has changed their current state of residence compared to the previous year. Interstate mobility is also measured by looking at long-term mobility with a difference of 5 years. The last variable in this section measures division mobility with a 5-year lag, with census division areas being large geographical agglomerates that encompass many

Table 3: Internal migration rates by skill level

	Less than HS	Some HS	HS graduate	Some college	College+
First gen: new migrants:					
N	1,149	2,187	7,134	3,327	11,342
State move	4.47	3.16	2.80	2.83	5.23
State move 5 yr	5.81	6.18	5.40	5.17	6.70
Division move 5 yr	2.77	3.81	3.93	3.02	4.67
First gen: early migrants					
N	3,387	8,047	23,653	11,812	29,617
State move	1.76	1.68	1.47	1.91	3.18
State move 5 yr	8.63	4.86	5.94	7.10	13.61
Division move 5 yr	7.68	3.83	3.74	4.01	11.10
First gen: old migrants					
N	4,916	12,713	43,295	30,340	49,934
State move	0.96	0.83	1.00	1.26	1.74
State move 5 yr	2.47	3.46	4.03	4.58	8.07
Division move 5 yr	2.02	2.35	2.63	2.48	5.92
Second gen					
N	179	3,264	25,121	29,063	44,764
State move	2.11	1.79	1.44	1.59	2.30
State move 5 yr	11.54	4.63	5.08	6.21	9.22
Division move 5 yr	0.00	3.38	3.64	4.31	6.87
Natives					
N	733	36,896	349,065	345,592	431,391
State move	1.67	1.92	1.56	1.72	2.38
State move 5 yr	5.38	5.85	5.93	6.93	9.88
Division move 5 yr	1.65	3.06	3.66	4.36	6.56

Note: Migration rates (in percentage points) by skill level and migrant generation.

Table 4: Internal migration rates by ethnicity

	White	Black	Asian	Mexican	Other Hispanic
First gen: new migrants					
N	4,743	2,163	7,743	6,835	6,478
State move	4.11	3.43	5.67	2.98	2.57
State move 5 yr	6.84	9.00	6.41	6.47	4.77
Division move 5 yr	5.06	5.92	4.85	4.02	3.26
First gen: early migrants					
N	12,807	6,474	22,530	25,758	18,928
State move	2.61	2.25	3.08	1.47	1.59
State move 5 yr	10.23	10.85	13.27	5.63	5.91
Division move 5 yr	7.36	7.48	11.20	3.91	4.07
First gen: old migrants					
N	29,307	10,046	36,817	51,309	29,196
State move	1.84	1.74	1.36	0.82	1.02
State move 5 yr	7.82	6.99	6.32	3.35	4.48
Division move 5 yr	5.17	4.58	4.87	2.16	3.12
Second generation					
N	41,586	3,890	15,419	29,158	12,446
State move	2.02	2.36	2.23	1.19	2.11
State move 5 yr	8.81	9.53	8.50	4.12	7.64
Division move 5 yr	6.29	6.45	6.94	3.12	5.40
Natives					
N	924,670	128,719	15,304	42,837	31,876
State move	1.95	1.84	1.62	1.34	2.06
State move 5 yr	7.93	7.40	7.52	5.12	7.53
Division move 5 yr	5.27	4.04	5.13	3.86	5.13

Note: Migration rates (in percentage points) by ethnicity and migrant generation.

states (Northeast, Southwest, etc.).

In the CPS data, migration history with the 5-year lag is available only for some specific waves: 1995, 2005, 2015, 2020, 2025. This causes a critical loss of about 1.3 million observations of respondents, which implies less information available for parameter estimation.

4.1 Baseline specification

The model is designed in this way:

$$Y_{ist} = \delta \cdot \text{migrant } gen_i + \beta \cdot X_i + \eta_s + \gamma_t + \epsilon_{ist} \quad (1)$$

Where Y_{ist} represents the mobility outcome, and X_i is a vector of controls that contains age, education, and gender. The model also absorbs state-specific characteristics that are time invariant and common state shocks in each specific year. Because observations are repeated within each family, standard errors are clustered at the household level (cpsid)³.

Regression results are presented in table 5.

Column 1 presents the estimation with state move as the dependent variable, column 2 with state move 5-year, and column 3 with division move 5-year.

The main coefficient of interest is the parameter that captures average differences in internal mobility for the three generations; the baseline is natives' mobility. Results highlight a larger probability in all the outcomes for first-generation migrants. Coefficients for the three columns range from 0.0047 to 0.0097, which translates into a 25% higher interstate mobility on average compared to natives⁴. Additionally, there are no differences between second-generation migrants and natives.

Estimates for the other controls in the model are consistent with established findings in the literature: years of education are positively correlated with the propensity to move for all individuals, males are more prone to move, while age is negatively correlated with mobility.

Overall, the direction of the estimates is consistent across each specification with proportional magnitudes, but sample size decreases for the specifications in columns 2 and 3. The differential is sharper for outcomes capturing longer distances (division move), suggesting first-generation mobility is concentrated in more substantive geographical moves.

³Results are robust to standard errors clustered at the state level.

⁴Changes are computed against the mean for natives.

In order to verify the robustness of these estimates, I have run the same specification in column 1 for the waves in which it is possible to compute the 5-year movement variables; results are displayed in appendix table 10.

I have also checked gender-specific differences, estimating the same model on a subset of male and female workers only. The results show that the differences are both less pronounced and less significant for females, while for the male cohort the coefficients' magnitude remains comparable with the baseline. Estimates from these models are in appendix tables 11 and 12.

Table 5: Reduced form estimates

Model:	State move (1)	State move (5 yr) (2)	Division move (5 yr) (3)
<i>Variables</i>			
First generation migrant	0.0047*** (0.0005)	0.0107*** (0.0021)	0.0093*** (0.0019)
Second generation migrant	-0.0001 (0.0006)	-0.0019 (0.0027)	-0.0003 (0.0026)
Years of education / 100	0.1782*** (0.0061)	0.7783*** (0.0289)	0.5938*** (0.0282)
Age / 100	-0.5510*** (0.0209)	-1.2304*** (0.0958)	-0.8071*** (0.0898)
Age squared / 100	0.0056*** (0.0003)	0.0112*** (0.0012)	0.0071*** (0.0011)
Male	0.0030*** (0.0002)	0.0078*** (0.0011)	0.0056*** (0.0011)
Mean dep. var. — natives	0.01857	0.07872	0.05
<i>Fit statistics</i>			
Observations	1,545,780	234,203	191,090
R ²	0.01022	0.03136	0.02173

Clustered (cpsid) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

All specifications include state and year fixed effects. Sample restricted to full-time workers aged 25–54 not enrolled in school, weighted by ASEC sample weights. Standard errors clustered at the household level (cpsid) in parentheses. Reference category is natives (3rd generation).

4.2 Heterogeneity analysis

In the previous section I have provided evidence that first-generation migrants display higher internal mobility probabilities compared with the other cohorts in the sample.

This pattern is consistent even after decomposing foreign-born migrants conditional on their arrival year in the US. I have re-estimated the model with new migrants

defined as first-generation migrants who arrived in the US no more than 5 years prior, early migrants who arrived between 5 and 15 years prior in the US, and old migrants with more than 15 years in the US.

The estimated parameters are displayed in figure 4. New migrants are the most mobile group overall, especially within the first generation. This result is not novel, as it was already described in previous papers (see Basso and Peri 2020; Borjas 2001; Maré, Morten, and Stillman 2007; and others). This fact is due to several factors. First, mobility across generations decreases monotonically as workers spend time in a specific community. Time invested in social ties and in the network substantially increases the trade costs of migration and resettlement. Second, new migrants are among those who usually have not yet developed a mature life cycle, both in terms of parental status and dwelling ownership. Estimating the conditional mobility with these specific demographic controls reveals a lower mobility for the first generation (next section).

Second-generation migrants are indistinguishable from natives on interstate moves, and the convergence hypothesis holds at this geographical scale.

5-year mobility displays a significantly higher level of uncertainty, reflected in bigger confidence intervals, due to sample size shrinkage. New migrants' coefficients for the 5-year mobility are reported but not reliable; in fact, they have low statistical significance.

Beyond this measurement issue, a substantive interpretation is that new migrants tend to settle in established enclaves where foreign-born individuals from the same nationality are residing; this phenomenon is driven specifically by asymmetric information on local labor market conditions before arrival and by the desire to leverage the network possibilities offered by a location with an already established enclave (David A. Jaeger 2007; Cadena 2014; Bartel 1989).

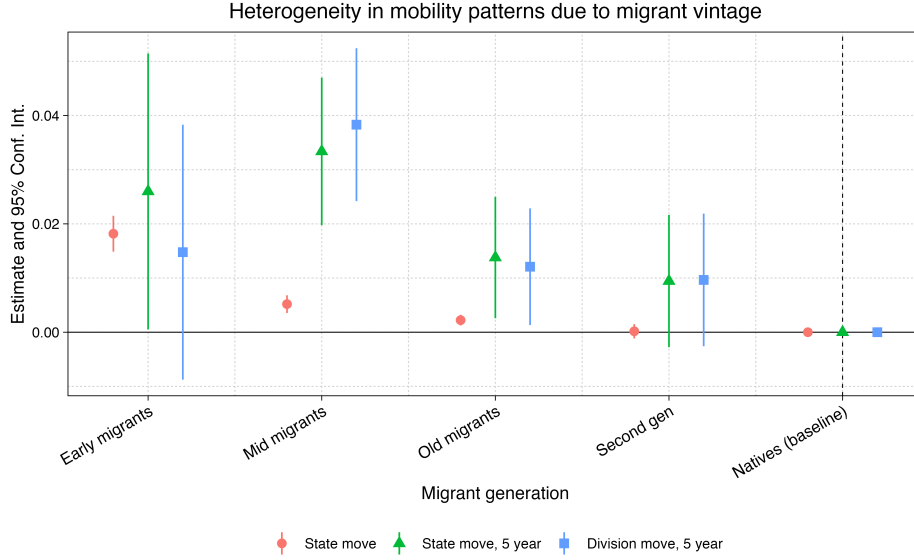
The model proposed in equation (1) does not allow us to uncover potential heterogeneity within each generation group. To do so I consider an alternative specification augmented with the interaction between migrant generation and two categorical variables: skill and ethnicity.

The extended model is defined as follows:

$$\begin{aligned}
Y_{ist} = & \alpha + \sum_{g \in \{1,2\}} \beta_g \mathbf{1}\{G_i = g\} + \sum_{k \neq k_0} \gamma_k \mathbf{1}\{H_i = k\} \\
& + \sum_{g \in \{1,2\}} \sum_{k \neq k_0} \delta_{gk} [\mathbf{1}\{G_i = g\} \times \mathbf{1}\{H_i = k\}] \\
& + X_i' \theta + \lambda_s + \tau_t + \varepsilon_{ist}.
\end{aligned} \tag{2}$$

Where the variable H_i represents either individual skill level or reported ethnicity.

Figure 4: Heterogeneity in mobility conditional to migrant vintage



Notes: the graph reports estimates and confidence intervals for the coefficients from an augmented version of equation (1). New migrants are foreign-born workers who have arrived in the country no more than 5 years prior. Early migrants have been residing in the country for 5 to 15 years, and old migrants have been in the US for more than 15 years. Standard errors are clustered at the household level, and the observations are weighted using CPS sample weights.

The baseline for generation is natives, "college or more" for skill and "Hispanic non-Mexican" for ethnicity.

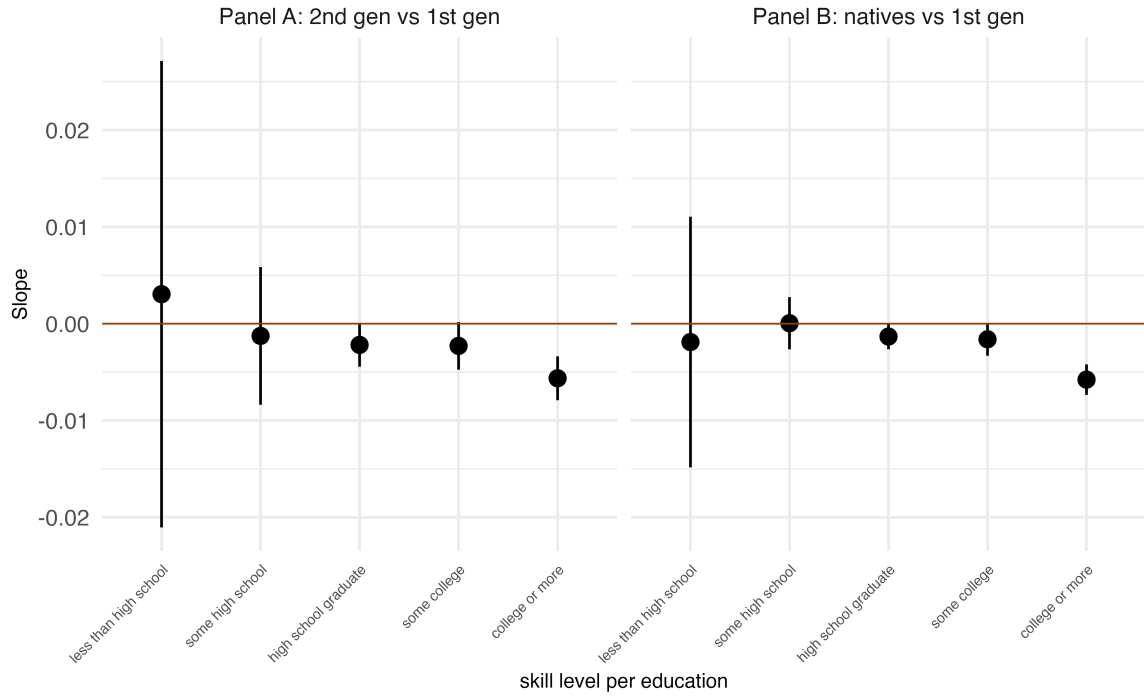
Regression results will be presented in terms of slope comparison and unconditional predictions. Full regression tables are available in appendix tables 13 and 15.

In figure 5 I have plotted the coefficient differences between the second generation and the first, and between natives and the first. These coefficients are taken from column 1 of the relevant table, as the outcome variable is state mobility in the short run.

Results show that, on average, foreign-born workers are more mobile compared to the second generation and natives in all the groups; the only two exceptions are "Less than high school" and "Some high school", in which the difference is not significant (partly due to the small size of the cell). Hence, first-generation migrants display a higher mobility overall, conditional on skill level. The first-generation group with college or more education displays the highest relative mobility.

Marginal slopes, however, tell us who moves more in relative terms, but not the unconditional mobility prediction. Figure 6 displays unconditional mobility predictions based on the models' estimates. Predictions are obtained by, for each (generation, skill) combination, assigning the entire estimation sample to that cell and averaging model-implied probabilities. Other covariates are held at their observed values.

Figure 5: Parameter comparison of heterogeneous skill mobility compared with foreign-born respondents

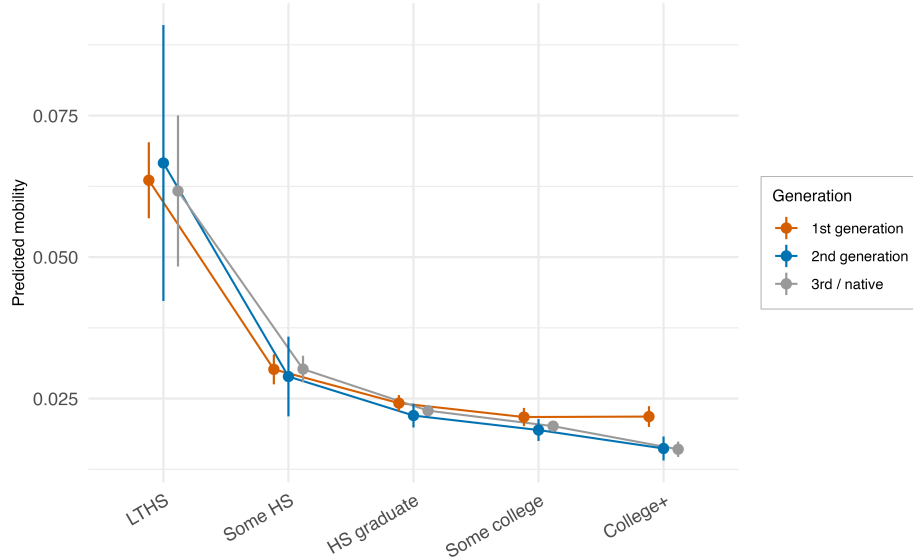


Notes: The figure reports the difference between the parameters from equation (2), estimated via OLS. Each column corresponds to one of the skill groups defined in the data. The panel on the left displays the difference between 2nd gen respondents and foreign-born individuals; the panel on the right displays the difference between natives and foreign-born individuals.

The fixed-effects structure absorbs the (state $\times$ ethnicity) and time dimensions.

It is possible to visualize the consistency of the marginal comparison from the previous plot; for example, within the "College or more" group, the first generation displays the highest predicted mobility. This picture conveys a clear message: the predicted mobility from the model is monotonically decreasing in skill.

Figure 6: Predicted mobility level by skill characterization.



Notes: the figure displays the predicted levels of mobility for different generations of migrants decomposing the prediction on the 5 skill levels per education. The prediction comes from the estimation of equation (2). SE are clustered at the household level.

Unobserved heterogeneity also emerges when looking at ethnicity differences. Figure 7 decomposes marginal mobility in the same spirit as for the skill variable. The estimates for the ethnicity model include state x ethnic fixed effects. Since the interaction of interest is generation x ethnicity, omitting state x ethnicity would let the interaction coefficients absorb state-specific ethnic patterns, for example, established Mexican networks in California or Asian communities in coastal metros. This concern is particularly relevant for southern states with large Mexican populations. For symmetry, Appendix Table 14 reports the skill model augmented with state x skill fixed effects; results are stable across the two specifications.

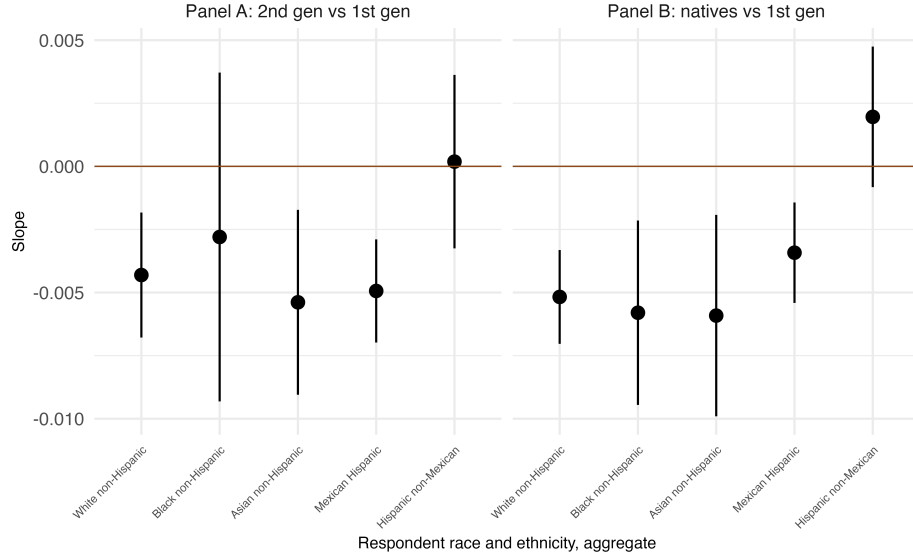
First-generation migrants are more mobile than the other groups across all ethnicities, except in the Hispanic non-Mexican and Black non-Hispanic categories, where the gap with the second generation is not significant.

A second dimension of heterogeneity comes from ethnicity. Figure 8 displays unconditional mobility rates along ethnicity. There are some interesting patterns that emerge from this picture.

First, the most mobile group is the Asian-first gen combination, confirming the rising importance of this cohort in US internal migration noted by Postel 2026. Identifying the mechanisms behind this pattern is beyond the scope of this paper.

Another interesting fact is the relatively low mobility of Mexican migrants (where the second generation displays the lowest point estimate). This lower level for naturalized Americans from that ancestry and for the second generation is important

Figure 7: Comparison of mobility across different ethnic groups.



Notes: the graphs display the marginal coefficient from equation (2) of each generation and ethnic pair. The points are the difference between the second generation (left) and natives (right) compared to first-generation migrants.

because of the relative share of Mexicans among the immigrant population in the US. This lower mobility may reflect, in part, the deterrent effect of migration enforcement programs (Secure Communities, 287(g)) that intensified during the sample period.

The heterogeneity analysis reveals two empirical regularities: a monotonic decline in mobility with skill, and a strong ethnic gradient where Asian first-generation migrants display the highest mobility and Mexican migrants the lowest across all generations.

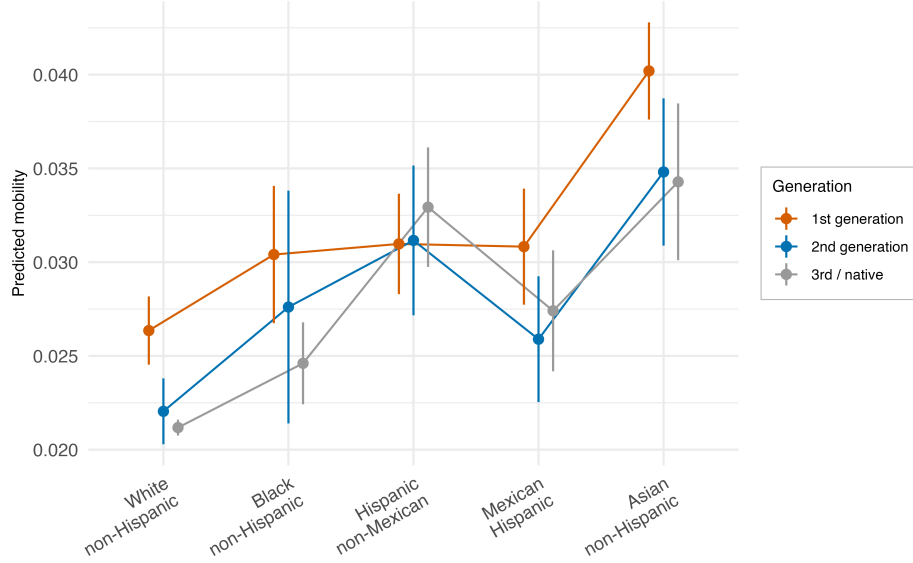
4.3 Accounting for life cycle stage and ownership

Results from the previous section show the differential across migrant generations, emphasizing the higher mobility for foreign-born workers compared to second-generation migrants and natives.

To address the potential importance of the workers' life cycle, in this section I have added a set of controls that can help me account for these differences by comparing individuals who are at similar life stages. The controls added proxy workers' life stage with the number of children living in the household, the number of children less than 5 years old, and a dummy variable equal to 1 if the respondent owns their residential unit.

Adding home-ownership as a control changes the estimand: the resulting coefficients measure the generation differential conditional on housing tenure, rather than the total generation differential. This is because home-ownership is plausibly an out-

Figure 8: Predicted level of mobility for different ethnic groups and migrant generations.



Notes: the graph reports the average level of mobility from the estimates of equation (2), with the interaction variable being the respondents' ethnicity. SE are obtained by clustering at the household level.

come of the same process that drives mobility (first-generation migrants own less because they are more mobile, or move less once they become owners). Thus, this specification should be interpreted as a decomposition of the housing channel rather than as a more credible estimate of the generation effect.

Results from this LMP specification are presented in table 6. Column 1 of table 5 and column 1 of table 6 are the same; column 2 adds the child controls, and column 3 adds ownership status.

The coefficient for 1-year interstate mobility of first-generation migrants slightly increases when adding household composition controls and decreases when controlling for ownership. Increasing the number of children in the household decreases the migration probability by 0.29 percentage points, and each additional child under 5 years old further decreases mobility by 0.09 percentage points.

The coefficient for ownership is negative and strongly significant. The first-generation-associated mobility predicted by the LPM is reduced by 70%, indicating that home-ownership accounts for the largest share of the gap between foreign-born workers and natives. Indeed, first-generation migrants are 16.5% more likely to own their dwelling, conditional on age and education (Appendix Table 17).

Still, after netting out the housing channel, a residual 7.5% differential remains for first-generation migrants, suggesting that home-ownership explains the bulk (but not all) of the baseline gap.

Table 6: Internal mobility conditional to life cycle stage

Model:	State move		
	(1)	(2)	(3)
<i>Variables</i>			
First generation migrant	0.0047*** (0.0005)	0.0053*** (0.0005)	0.0014*** (0.0005)
Second generation migrant	-0.0001 (0.0006)	-0.0002 (0.0006)	0.0002 (0.0006)
Years of education / 100	0.1782*** (0.0061)	0.1726*** (0.0061)	0.2401*** (0.0063)
Age / 100	-0.5510*** (0.0209)	-0.4414*** (0.0215)	-0.3416*** (0.0212)
Age squared / 100	0.0056*** (0.0003)	0.0043*** (0.0003)	0.0035*** (0.0003)
Male	0.0030*** (0.0002)	0.0030*** (0.0002)	0.0035*** (0.0002)
N. of children in hh		-0.0029*** (0.0001)	-0.0018*** (0.0001)
N. of children <5 yrs old in hh		-0.0009** (0.0004)	0.0000 (0.0004)
Owner			-0.0306*** (0.0005)
Mean dep. var. — natives	0.01857	0.01857	0.01857
<i>Fit statistics</i>			
Observations	1,545,780	1,545,780	1,545,780
R ²	0.01022	0.01087	0.02024

Clustered (cpsid) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Note: Dependent variable is 1-year state move in all columns. Column 1 reproduces the baseline specification (Table 5, column 1); column 2 adds the number of children in the household and the number of children under 5; column 3 further adds an indicator equal to 1 if the respondent owns their residence. All specifications include state and year fixed effects; standard errors clustered at the household level.

4.4 Spatial decomposition

The results so far identify generational differences in mobility at the interstate scale. A natural question is whether these patterns hold at a finer scale, across counties within a state or within a county. The CPS March supplement provides variables that allow this decomposition.

The variable "migsta1" reports migration status with respect to one year prior. Unfortunately, the data do not provide any information on the past county or metropolitan area (the one from the year before), but among the codes that are available, there is one specific code that indicates whether the respondent made a within-county movement or an across-county, within-state movement.

This added layer of spatial decomposition allows me to complement the LPM estimates with these two proxies for short and medium movements, respectively⁵. Results from this geographical disaggregation are provided in table 7. The specification follows Section 4.3, including life-cycle and home-ownership controls.

The convergence pattern observed at the interstate scale (Section 4.2) does not hold at finer geographies. Second-generation migrants display significantly lower mobility than natives both within counties (-8.4%) and across counties within a state (-25.5%). The first generation, by contrast, is indistinguishable from natives within counties and less mobile across counties, suggesting that the higher interstate mobility documented in Section 4.1 reflects a preference for long-distance moves rather than higher mobility tout court.

This pattern is consistent with first-generation mobility being driven by destination-specific opportunities: when they move, they target distant locations with expected returns rather than reshuffling within a local area. Once they settle in a metropolitan area, they tend to stay.

To the best of my knowledge, this is the first paper to document systematically the lower within-state mobility of the second generation relative to natives in the contemporary US, using a long CPS panel that allows clean identification of generational categories.

This result is inconsistent with full assimilation to native mobility patterns: second-generation migrants appear more locally anchored than natives, possibly reflecting persistent ties to the ethnic enclaves established by their parents.

Taken together with the interstate results from Section 4.1, the spatial decomposition reveals a generational signature in the scaling of mobility: first-generation migrants move long but not short, while second-generation migrants move less than

⁵Because the CPS does not record the prior county of residence, I cannot identify origin-destination pairs for within-state moves, and standard corrections for spatial autocorrelation are not feasible.

Table 7: Medium and short distances internal mobility estimates.

Model:	Within county (1)	Within state (2)
<i>Variables</i>		
First generation migrant	-0.0002 (0.0010)	-0.0059*** (0.0006)
Second generation migrant	-0.0064*** (0.0013)	-0.0059*** (0.0007)
Years of education / 100	-0.0003 (0.0125)	0.1225*** (0.0071)
Age / 100	-1.3357*** (0.0431)	-0.5538*** (0.0271)
Age squared / 100	0.0128*** (0.0005)	0.0056*** (0.0003)
Male	0.0006 (0.0005)	0.0018*** (0.0003)
N. of children in hh	-0.0043*** (0.0003)	-0.0028*** (0.0002)
N. of children <5 yrs old in hh	0.0057*** (0.0008)	0.0015*** (0.0005)
Owner	-0.0998*** (0.0009)	-0.0289*** (0.0006)
Mean dep. var. — natives	0.07597	0.02314
<i>Fit statistics</i>		
Observations	1,545,780	1,545,780
R ²	0.06819	0.02329

Clustered (cpsid) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Note: Within county equals 1 if the respondent's prior-year move was within the same county; within state equals 1 if the move was across counties but within the same state. Both are coded 0 if no move or an interstate move occurred, and missing for respondents with no valid prior-state migration information (not in universe or residing abroad). Specifications include the life-cycle and home-ownership controls of Section 4.3 (number of children, number of children under 5, ownership) plus state and year fixed effects; standard errors clustered at the household level.

natives at all scales finer than the state. The mechanism behind this asymmetry, whether driven by enclave attachment, housing tenure or selection, is left for future work.

5 Gravity model of internal mobility⁶

The reduced-form estimates presented in the previous section describe whether individuals belonging to different migrant generations display different probabilities of moving across U.S. states. They do not, however, describe which destinations are selected by movers, nor which origin-destination characteristics explain the allocation of migration flows. The gravity model explores this second margin by moving from an individual-level mobility indicator to a corridor-level flow framework.

This section is structured in the following way: first, I will provide a remark on the foundations of gravity models, deriving the PPML estimation procedure from an individual indirect utility maximization framework; subsequently, I will describe the estimating equation brought to the data; and finally, I will discuss the estimation results.

5.1 Theoretical framework

This section derives the estimating equation from a random utility maximization (RUM) framework in the spirit of McFadden 1974, following the gravity implementation in Beine, Bertoli, and Fernández-Huertas Moraga 2016. The derivation motivates the use of the Poisson Pseudo-Maximum Likelihood (PPML) estimator (Silva and Tenreyro 2006) and clarifies which sources of variation identify the parameters of interest.

Consider a worker i residing in origin state o at time t and belonging to migrant generation g . Their utility function can be represented as:

$$U_{iodtg} = V_{odtg} + \varepsilon_{iodtg}$$

The deterministic part of the utility function V_{odtg} takes the form:

$$V_{odtg} = D'_{d,t-2}\theta_g + W'_{od}\beta_g + Net_{od,t-2}\rho_g$$

where $D_{d,t-2}$ are the destination-specific pull factors, W_{od} is a vector of dyadic-specific characteristics, and $Net_{od,t-2}$ proxies the ethnic strength of the corridor. Des-

⁶This section is a work in progress. The empirical results are based on the current specification of the corridor variables; identification of generation-specific effects of the ethnic corridor and the construction of valid standard errors following Adão, Kolesár, and Morales 2019 are ongoing.

tion and network characteristics enter with a two-year lag to reflect the information set available to potential movers at the migration decision date and to mitigate concerns of simultaneity between contemporaneous flows and destination characteristics.

Assuming that $\varepsilon_{i\text{od}tg}$ follows a Type-I Extreme Value distribution, the probability of a migrant choosing destination d can be written as:

$$P(d \mid o, i, t, g) = \frac{\exp(D'_{d,t-2}\theta_g + W'_{od}\beta_g + Net_{od,t-2}\rho_g)}{\sum_{s \in \mathcal{D}_o} \exp(D'_{s,t-2}\theta_g + W'_{os}\beta_g + Net_{os,t-2}\rho_g)}.$$

Let M_{otg} denote the number of potential movers from origin (o), generation (g) in year (t). The expected bilateral flows are given by:

$$\mathbb{E}[Flow_{odtg}] = M_{otg}P_{odtg}.$$

Substituting the conditional logit probability P_{odtg} this yields:

$$\mathbb{E}[Flow_{odtg}] = \frac{M_{otg}}{\Omega_{otg}} \exp(D'_{d,t-2}\theta_g + W'_{od}\beta_g + Net_{od,t-2}\rho_g),$$

where

$$\Omega_{otg} \equiv \sum_{s \in \mathcal{D}_o} \exp(D'_{s,t-2}\theta_g + W'_{os}\beta_g + Net_{os,t-2}\rho_g).$$

Since $\ln M_{otg} - \ln \Omega_{otg}$ varies only at the origin-year-generation level, it is absorbed by origin-year-generation fixed effects. The final estimating equation is the following:

$$\mathbb{E}[Y_{odtg} \mid X_{odtg}] = \exp(\alpha_{otg} + D'_{d,t-2}\theta_g + W'_{od}\beta_g + Net_{od,t-2}\rho_g). \quad (3)$$

Where the parameters are estimated with PPML. The conditional mean is log-linear in the parameters, which would suggest OLS estimation of $\ln(Y_{odtg})$. However, Silva and Tenreiro 2006 show that in the presence of heteroskedasticity the log-linearization introduces inconsistency through Jensen's inequality. PPML consistently estimates the parameters of the conditional mean under weaker assumptions, requiring only correct specification of the conditional expectation rather than a Poisson data-generating process. A further practical advantage is that PPML naturally accommodates observations with zero flows, which are non-trivial in the origin-destination-year-generation panel and would be dropped under a log transformation. The parameters of interest $\Theta = \{\theta_g; \beta_g; \rho_g\}$ are identified from variation in the data at the destination within origin-year-generation cells. Section 5.3 introduces this baseline specification; Sections 5.4 and 5.5 progressively saturate the fixed-effect structure with destination-year-generation and origin-destination terms to isolate the ethnic corridor

channel.

5.2 Bartik enclave measure

The relevance of networks in migration decisions is well documented in the migration literature. To account for the network effect at the corridor level, I create an ethnic measure that combines the network similarity across each origin and destination pair. The variable $Net_{od,t-2}$ is a measure of overlap between state birthplace compositions.

The stock of co-ethnics currently residing in a state is the outcome of past migration decisions responding to the same economic drivers that shape contemporaneous flows. Using this stock directly would confound the network channel with unobserved destination-specific shocks correlated with the network itself.

Following the shift-share literature (Card 2001), I construct a predicted composition measure based on historical allocation shares and national shifts, using 2000 as the base year and a leave-one-out construction to mitigate the mechanical reflection of the state’s own contribution to the national total.⁷

Let $b \in \mathcal{B}$ be the birthplace of worker i , s the state, and $\tau = 2000$ the base year. The choice of 2000 as the base year reflects a trade-off: shares constructed at the year 2000 are sufficiently predetermined relative to the estimation period (from 2002 to 2025), but at the same time they retain predictive power for the current state composition.

The historical share of birthplace group b located in state s in base year τ is defined as:

$$\pi_{bs\tau}^{\text{base}} = \frac{pop_{bs\tau}}{\sum_{s' \in \mathcal{S}} pop_{bs'\tau}}.$$

Let N_{bt} denote the national population of birthplace group b , and let pop_{bst} denote the population of group b in state s . I construct a leave-one-out national total:

$$N_{bst}^{LOO} = N_{bt} - pop_{bst}$$

which will be used to predict the state composition net of the endogenous state contribution to the national stock.

The predicted population of birthplace group b in state s , based on the historical allocation shares, is then:

$$\widehat{pop}_{bst}^{\tau} = \pi_{bs\tau}^{\text{base}} \cdot N_{bst}^{LOO}.$$

⁷Section 5.5 discusses sensitivity to the choice of the base year.

Normalizing across birthplace groups gives the predicted birthplace-composition share:

$$\hat{\pi}_{sbt}^{\tau} = \frac{\widehat{pop}_{bst}^{\tau}}{\sum_{b' \in \mathcal{B}} \widehat{pop}_{b'st}^{\tau}}.$$

Finally, I denote the shift-share corridor network between origin o and destination d as the overlap between their predicted birthplace composition vectors:

$$SSNet_{odt}^{\tau} = \sum_{b \in \mathcal{B}} \hat{\pi}_{obt-2}^{\tau} \cdot \hat{\pi}_{dbt-2}^{\tau}$$

which is the inner product of predicted composition shares. The interpretation is the following: the higher the historical similarity in birthplace composition between origin and destination, the higher the value of $SSNet_{odt}^{\tau}$. If the two composition vectors are orthogonal, the value approaches 0. The inner product weights each birthplace group by the product of its shares in the two states, so overlap on numerically important groups contributes more than overlap on small groups.

The 2000 Census 5% sample provides the historical allocation shares $\pi_{b\sigma\tau}^{base}$ used in the baseline construction. Earlier decennial census 5% samples (1900, 1910, ..., 1990) are used in Section 5.5 to assess sensitivity to the choice of base year. Birthplace groups $\mathcal{B}$ are defined at the country-of-birth level, using IPUMS harmonized birthplace codes.

This aggregate measure does not distinguish between the network of first-generation co-ethnics currently residing at destination and the historical enclave built by second-generation migrants born to earlier waves. Section 5.4 proposes a decomposition of these two channels using generation-specific shift-share measures constructed from CPS data at a coarser (11 macro-region) origin aggregation, dictated by CPS sample size constraints on parental birthplace.

5.3 Baseline gravity estimates

Recall the estimating equation for the gravity model derived in Section 5.1, equation (3). The vector of state-specific pull factors $D_{d,t-2}$ contains a series of variables that describe states' characteristics: population, share of graduates, employment rate, log average wage, and share of migrants. These variables are used to predict destinations' attractiveness for migration flows along these channels. Dyadic characteristics W_{od} contains specific origin-destination characteristics such as distance in log and adjacency.

The network variable $Net_{od,t-2}$ in equation (3) is built as two separate shift-share measures, both constructed following the procedure described in Section 5.2, but differing in the definition of the birthplace set $\mathcal{B}$. The foreign enclave measure $SSNet_{odt}^{for}$

takes $\mathcal{B}$ as the set of countries of birth outside the US and captures the historical similarity of immigrant composition between origin and destination states.

In the model I have included two variables: the first proxies internal migration corridors of the native population and is constructed with US states as workers' birthplace. The second is constructed on immigrant workers' birthplaces. Both variables are rescaled by their standard deviation.

Estimates are presented in table 8.

Table 8: Gravity model estimates

Dependent Variable:	flow			
Model:	Baseline (1)	Multilateral (2)	Heterogeneous (3)	Corridor (4)
<i>Variables</i>				
Log distance	-0.5619*** (0.0482)	-0.5956*** (0.0460)	-0.6400*** (0.0503)	
Border	0.6915*** (0.0977)	0.5430*** (0.0689)	0.5102*** (0.0712)	
Foreign enclave (Bartik)	0.0636** (0.0321)	-0.0561* (0.0328)	-0.0706** (0.0354)	-0.3976 (0.2980)
Internal enclave (Bartik)	0.1499*** (0.0275)	0.2514*** (0.0235)	0.2630*** (0.0260)	-0.5212*** (0.1387)
Population / 1000	0.0002*** (0.0000)			
Share of graduates	1.7191*** (0.6194)			
Employment rate	1.8723 (1.7700)			
Log average wage	-0.6488* (0.3378)			
Share of migrants / population	0.3993 (0.5470)			
Log distance - 1st gen			0.1525** (0.0680)	
Log distance - 2nd gen			0.2496** (0.1038)	
Border - 1st gen			0.0788 (0.1056)	
Border - 2nd gen			0.1101 (0.1421)	
Foreign enclave - 1st gen			0.0812 (0.0560)	0.0590 (0.0538)
Foreign enclave - 2nd gen			0.0539 (0.0719)	0.0279 (0.0747)
Internal enclave - 1st gen			-0.0493 (0.0307)	-0.0292 (0.0196)
Internal enclave - 2nd gen			0.0135 (0.0455)	0.0088 (0.0297)
<i>Fixed-effects</i>				
origin-year-generation	Yes	Yes	Yes	Yes
destination-year-generation		Yes	Yes	Yes
origin-destination				Yes
<i>Fit statistics</i>				
Observations	136,250	114,421	114,421	93,916

Clustered (od) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Note: Dependent variable is bilateral migration flow Y_{odtg} , estimated via PPML following equation (3). All columns include origin-year-generation fixed effects. Column 2 adds destination-year-generation fixed effects. Column 3 interacts the dyadic variables and the two Bartik enclave measures with first- and second-generation indicators; reported coefficients are deviations from the native baseline. Column 4 adds origin-destination fixed effects, which absorbs the time-invariant state-level pull factors reported in column 1. Foreign and internal enclave measures are rescaled by their standard deviation. Standard errors clustered at the origin-destination pair level in parentheses.

The fixed-effect structure gradually saturates the model, absorbing different sources of variability. All the specifications include (origin, year, generation) fixed effects. Column 2 estimates the impact of dyadic variables and network on migration flows in a multilateral resistance model, with the addition of (destination, year, generation) fixed effects. Column 4 concludes with the inclusion of (origin, destination) fixed effects, where the only source of variation left for identification is within each corridor at the (year, generation) level.

Distances are negatively correlated with migration flows in the US between 2002 and 2025. A 1% increase in the logarithm of distance in km is associated with a -0.56% decrease in average migration flows, which is consistent with previous findings in the literature. Also, adjacent origin-destination pairs are associated with higher average migration flows compared with non-adjacent states. These results are robust when absorbing state-year specific characteristics in column 2.

Population and share of graduates are positive predictors of migration flows. A unit increase in each variable (consistent with its scale) leads to 0.020% and 1.73% higher migration flows, respectively.

The coefficient on log average wage is negative and marginally significant. This apparent anomaly likely reflects a compensating differential mechanism: higher-wage destinations are also higher-cost destinations, and net rather than gross earnings drive location decisions. Consistent with this interpretation, Ganong and Shoag 2017 document a decline in migration flows towards high-wage, high-cost states in recent decades. Appendix Table 16 reports a specification that includes log average rent as an additional control; the wage coefficient becomes closer to zero, supporting the interpretation above.

Column 1 shows that both network measures are strongly and positively associated with migration flows when the specification treats them as destination pull factors alongside other state characteristics. A one standard-deviation increase in the foreign enclave measure is associated with a 6.6% higher flow, and a comparable increase in the internal enclave with a 16.2% higher flow. Column 2 introduces destination-year-generation fixed effects, absorbing any time-varying destination characteristics common across origins. Under this specification the coefficient on the foreign enclave changes sign and becomes marginally significant, indicating that the positive coefficient in Column 1 was partly capturing unobserved destination-side factors correlated with the ethnic composition of the state — for example, industry mix or labor demand shocks that plausibly attract new migrants and are also correlated with historical immigrant enclaves. The internal enclave coefficient instead remains positive and grows in magnitude, suggesting that the native-corridor channel does not suffer from the same omitted-variable bias (OVB). Column 4 further saturates the specifica-

tion with origin-destination fixed effects, restricting identification to within-corridor variation over time. Under this most stringent specification, both network measures turn negative, with the internal enclave becoming strongly significant. As I discuss in Section 5.5, this reversal is diagnostic of the serial correlation problem highlighted in David A Jaeger, Ruist, and Stuhler 2018: shift-share measures based on distant historical shares partly capture long-run persistence in migration patterns rather than a well-identified network channel. Column 3 introduces generation-specific interactions on the dyadic variables. First and second-generation migrants exhibit a significantly lower elasticity of flows to distance than natives (deviation coefficients 0.15 and 0.25, respectively, both significant at conventional levels). This is a first indication that gravity fundamentals themselves operate differently across generations, motivating the more systematic decomposition proposed in Section 5.4.

This baseline specification treats the ethnic network as a single aggregate channel, common to all migrant generations. But the theoretical prediction is asymmetric: first-generation migrants should respond more strongly to the presence of co-ethnic communities at destination, while natives should respond primarily to the strength of native internal-migration corridors. Section 5.4 tests this hypothesis by constructing generation-specific shift-share measures and estimating their differential effect across generations.

5.4 Generation-specific ethnic corridors

This section documents the key empirical finding of the gravity analysis: first-generation migrants respond specifically to co-ethnic enclaves historically built by second-generation communities, a channel that is silent for both second-generation migrants and natives. This asymmetry is uncovered by decomposing the aggregate network measure of Section 5.3 into three generation-specific shift-share variables that isolate the relevant network for each mover type.

I create generation-specific enclave measures following the methodology described in Section 5.2. The relevant network for first-generation migrants is the composition of foreign-born workers in each state; the network for the second generation corresponds to the parental network, measured with the composition in each state conditional on migrants' parental origin. For native movers, the relevant network remains the same as the previous model in Section 5.3, measured by internal corridors conditional on workers' state of birth.

The source of data used is a mixture of Census decennial data (5% from 1900 to 2000), ACS (1% from 2001 to 2025), and CPS ASEC. The usage of this mixture of sources is mandated by the specification itself: ACS data does not contain any

information on parental background, as discussed in Section 3. Additionally, I have grouped foreign states of origin into 11 macro-regions: Northern and Western Europe, Southern Europe, Eastern Europe and the USSR area, Mexico, Central America and the Caribbean, South America, East Asia, South and Southeast Asia, the Middle East and North Africa (MENA), and other. The 11-macro-region aggregation is dictated by CPS sample size at the parental-birthplace level. A finer aggregation (country of birth, as used in Section 5.2 with Census data) would yield share estimates with prohibitive noise for the second-generation subsample, where each cell must be populated with respondents reporting parental birthplace at that granularity.

The first-generation network-specific measure has been constructed with $\mathcal{B}$ corresponding to workers' region of origin from CPS data. Second-generation network is constructed starting from parents' region of origin. The population has been restricted in order to account for only one region of origin (both parents from the same region or one parent born abroad and the other in the US). Workers with parents from different birthplaces are dropped from the analysis.⁸

The baseline shares are computed by pooling CPS ASEC waves from 1995 to 2000. Pooling is necessary because CPS sample sizes at the (parental birthplace $\times$ state) level are too small in any single year to produce reliable share estimates, especially for the second-generation subsample. The trade-off is between statistical precision of the shares and their pre-determination relative to the estimation period (2002–2025). Because the demographic composition of birthplace groups within states evolves slowly (year-on-year correlations exceed 0.95), the pooling window introduces limited attenuation of the pre-determination argument while substantially reducing measurement error in the shares.

Native network is measured in the same way as in Section 5.3, with baseline year 2000.

Table 9 presents the estimates from this model.

⁸They account for roughly 7% of the total, which should not meaningfully impact the estimates' magnitude.

Table 9: Gravity model estimates with generation-specific Bartik enclave measures

Dependent Variable:	flow			
Model:	Baseline (1)	Multilateral (2)	Heterogeneity (3)	Corridor (4)
<i>Variables</i>				
Logarithm of distance	-0.5638*** (0.0645)	-0.5730*** (0.0688)	-0.5713*** (0.0681)	
Border	0.6907*** (0.1008)	0.5564*** (0.0795)	0.5482*** (0.0785)	
Bartik internal structure	0.1492*** (0.0329)	0.2486*** (0.0301)	0.2738*** (0.0312)	-0.5447*** (0.1358)
Bartik 1st gen enclave	0.1188*** (0.0303)	0.0374 (0.0393)	0.0510 (0.0435)	-0.0145 (0.0498)
Bartik 2nd gen enclave	-0.0200 (0.0236)	-0.0224 (0.0172)	-0.0469** (0.0182)	0.0131 (0.0200)
Population / 1000	0.0002*** (0.0000)			
Share of graduates	1.6524*** (0.4495)			
Employment rate	1.7522 (1.7727)			
Log average wage	-0.6232* (0.3749)			
Share of migrants / population	0.2619 (0.4582)			
Bartik internal structure - 1st gen			-0.0891*** (0.0182)	-0.0278* (0.0142)
Bartik internal structure - 2nd gen			-0.0675** (0.0286)	0.0076 (0.0269)
Bartik 1st gen enclave - 1st gen			-0.0394 (0.0816)	-0.0085 (0.0864)
Bartik 1st gen enclave - 2nd gen			-0.0096 (0.0734)	0.0043 (0.0725)
Bartik 2nd gen enclave - 1st gen			0.0886*** (0.0294)	0.0646** (0.0289)
Bartik 2nd gen enclave - 2nd gen			0.0344 (0.0292)	0.0299 (0.0284)
<i>Fixed-effects</i>				
origin-year-generation	Yes	Yes	Yes	Yes
destination-year-generation		Yes	Yes	Yes
origin-destination				Yes
<i>Fit statistics</i>				
Observations	136,250	114,421	114,421	93,916

Clustered (origin) standard-errors in parentheses

Signif. Codes: ***, 0.01, **, 0.05, *, 0.1

Note: Dependent variable is bilateral migration flow, estimated via PPML. The aggregate network measure of Table 8 is decomposed into three generation-specific shift-share measures: the native internal-corridor measure (Bartik internal structure), the first-generation foreign-enclave measure (Bartik 1st gen enclave), and the second-generation parental-enclave measure (Bartik 2nd gen enclave), all constructed following the methodology of Section 5.2. Column 1 includes origin-year-generation fixed effects. Columns 2 and 3 add destination-year-generation fixed effects; Column 3 further interacts each network measure with first- and second-generation indicators, with reported interaction coefficients expressed as deviations from the native baseline. Column 4 keeps these interactions and further adds origin-destination fixed effects. Standard errors clustered at the origin level in parentheses.

Destination pull factors in column 1 remain stable compared to the previous specification both in sign and magnitude. The network-specific variables show a positive and significant estimate for the native and first-generation networks. The second-generation network coefficient is not a significant predictor of migration flows.

Column 2, with the additional destination-year-generation fixed effects, reveals a consistent pattern with Section 5.3. The second-generation parameter remains insignificant, first generation loses its predictive power after netting out destination-specific characteristics, and the internal enclave measure maintains the same direction and magnitude.

The interaction specifications in Columns 3 and 4 reveal three distinct patterns. First, the native internal enclave attracts native movers significantly more than either first- or second-generation migrants. The deviation coefficients on the interaction with the first and second generations are both negative and significant (-0.089 and -0.068 in Column 3, respectively), confirming that the internal-corridor channel is largely a native phenomenon. Second, first-generation migrants show no additional responsiveness to the network of their own generation, above the common baseline: the interaction between the first-generation shift-share and the first-generation indicator is insignificant. Third, first-generation migrants exhibit a statistically significant additional responsiveness to the historical enclave built by second-generation communities. The interaction coefficient is 0.066 (s.e. 0.023 , $p < 0.01$) in Column 3 and remains significant at 0.065 (s.e. 0.029) under the more stringent origin-destination specification of Column 4. Second-generation migrants show no analogous deviation on this channel. Taken together, these three patterns paint an economically coherent picture. New arrivals do not draw on the network of peers arriving alongside them, which lacks established community infrastructure. They draw on the mature enclaves built by the second generation — the US-born children of earlier immigrant waves — which offer decades of accumulated informational and social capital.

The exclusively first-generation nature of the parental-enclave channel is the central empirical contribution of the gravity analysis. New arrivals rely on the co-ethnic community infrastructure built by earlier waves — but this reliance is invisible when the network is measured as a single aggregate, as in Section 5.3, and requires the generation-specific decomposition proposed here to be identified.

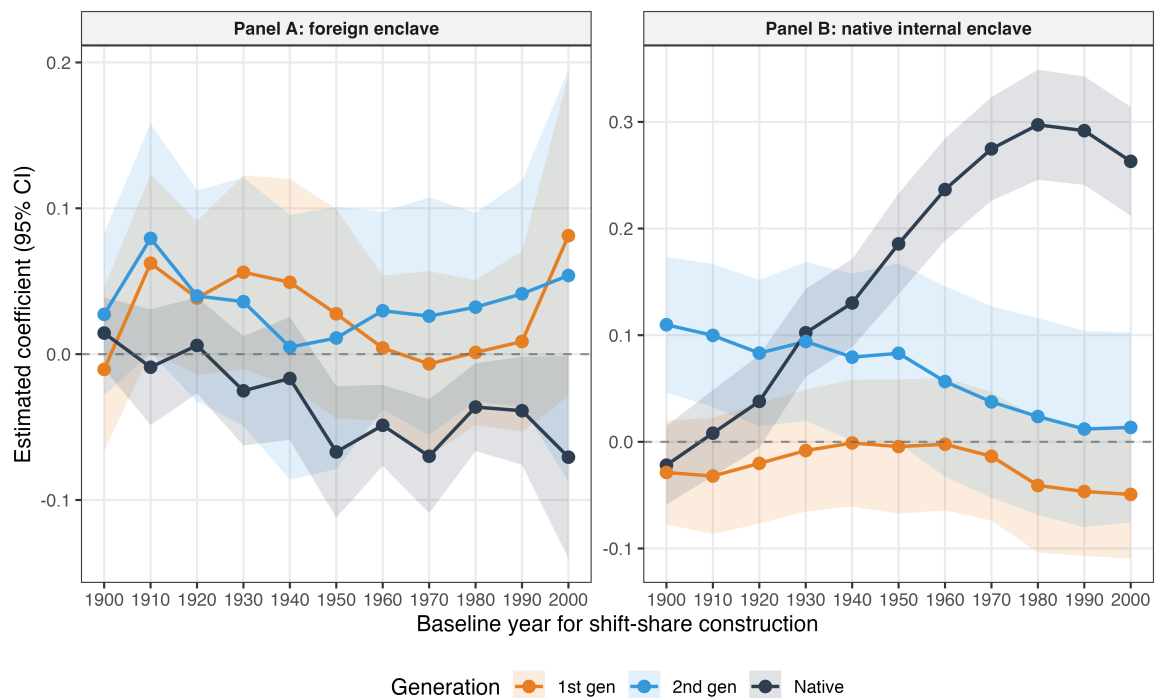
The sensitivity analysis in Section 5.5 is confined to the native internal enclave measure of Section 5.3, for which decennial Census data allow the base year to be pushed back to 1900. The generation-specific shift-shares of the present section cannot be constructed with pre-1995 shares because CPS parental birthplace information is unavailable before that date.

5.5 Sensitivity to the base year

The shift-share variables are likely mixing long-run and short-run effects of state demographic composition on migration flows, as framed by David A Jaeger, Ruist, and Stuhler 2018. To test for this issue in the data, I have repeatedly computed the gravity model in Section 5.3, column 3, with different baselines for the shares. If nearer baselines compared with the estimation period have higher predictive power, then the variable is likely mixing the short-run prediction of the composition with the associated long-run effect of the share construction. On the other hand, a constant coefficient throughout the baseline range would imply that the ethnic channel is not dependent on the baseline decision.

Figure 9 displays the pattern of the coefficients for both measures, the foreign and internal network.

Figure 9: Bartik measure sensitivity to year baseline



Note: The graph reports, for the specification of Table 8, column 3, the point estimate and confidence interval of the foreign- and internal-enclave Bartik coefficients when the shift-share measures are re-constructed using historical decennial Census allocation shares from 1900 to 2000 (10-year intervals) as the base year, holding the rest of the specification fixed. Standard errors clustered at the origin-destination pair level.

The x-axis displays the different baseline years, from 1900 to 2000. The y-axis displays the coefficient point estimate and confidence interval.

The graphs show two distinct patterns. The Bartik network for the native composition is increasing as the baseline year is nearer the sample of analysis, while the

coefficients for the first and second generations are constant over time. Panel A displays a stable pattern near zero across all baselines and years.

The chart suggests that the Bartik measure for the native corridor is likely capturing a source of serial correlation, which identifies the persistence of the corridor. Once the demographic composition of the corridor has been established, it remains stable over time. On the other hand, the network of foreign-born individuals is robust to different baselines. The interpretation for this evidence is that the foreign-born composition of migrants has changed over time, consistent with migration shocks that caused different sources of immigration in the US.

The internal enclave measure in column 4 should be interpreted with caution because there is sufficient evidence that suggests it is capturing a mixture of serial correlation and short-run effects on internal migration flows. On the other hand, the graphs provide a source of robustness for the findings in Section 5.4, which rely on a different data source but likewise pertain to the foreign channel.

Finally, standard inference in shift-share designs is complicated by the fact that observations sharing similar exposure shares are not independent (Adão, Kolesár, and Morales 2019). The AKM correction requires bootstrapping the share component to retrieve the empirical distribution function of the parameters; however, in my setup this is unfeasible due to the 11 macro-regions and their specific contribution to the enclave measure. As an alternative, I cluster standard errors at the origin level. The parental-enclave finding of Section 5.4 remains statistically significant under this clustering ($\beta = 0.066$, s.e. = 0.030, $p = 0.028$).

6 Conclusion

In this paper I have studied internal mobility patterns for different migrant generations. The analysis included many dimensions with different internal migration outcomes, both at the state level and at a finer geographical level. In interstate mobility, first-generation migrants display higher internal mobility compared to natives. The first-generation premium reverses sign at finer scales: they show no within-county differential and are about 25% less mobile across counties within a state. The same pattern does not hold within counties, as they show the same mobility rates as natives. This is consistent with first-generation mobility being driven by long-range wage and labor-market differentials, while local networks anchor them once they settle in a metropolitan area.

Second-generation migrants behave similarly to natives in interstate mobility, signaling that at this geographical scale there is convergence in patterns. When considering the within- and across-county scale instead, this group reveals significantly lower

mobility rates, behaving differently from natives. These new results seem to suggest that family ties of the second generation are more binding for individuals of this type, who benefit from both family and community ethnic networks.

Decomposition of the unobserved heterogeneity signals the relevance of interstate mobility rates for the Asian ethnic group, a recent finding in the literature that emphasizes their role. Mexican mobility, instead, seems to be slower, especially for the first and second generations.

The skill decomposition reveals the premium for the most educated first-generation migrants, who are the most mobile across all the skill and generation groups.

Controlling for home ownership reduces the first-generation premium by approximately 70%, identifying housing tenure as the main channel through which the gap operates. A residual difference of about 7.5% remains, indicating that home ownership alone does not fully account for the higher mobility of foreign-born workers.

Preliminary gravity estimates suggest a positive role of corridor similarity in shaping bilateral flows, with heterogeneous magnitudes across generations. A full discussion of these results is in Section 5; the identification of generation-specific effects of the ethnic corridor is left to ongoing work.

The absence of within-state origin codes in the CPS limits the identification of the mechanisms behind these patterns. Yet the empirical regularity is robust: second-generation migrants do not behave as natives at finer scales.

Aggregating them with the native population, as the existing literature typically does, overestimates domestic mobility and masks a generational anchoring effect that may matter for models of local labor market adjustment.

References

- Adão, Rodrigo, Michal Kolesár, and Eduardo Morales (Nov. 2019). “Shift-Share Designs: Theory and Inference*”. en. In: *The Quarterly Journal of Economics* 134.4, pp. 1949–2010. ISSN: 0033-5533, 1531-4650. DOI: [10.1093/qje/qjz025](https://academic.oup.com/qje/article/134/4/1949/5552146). URL: <https://academic.oup.com/qje/article/134/4/1949/5552146> (visited on 05/30/2026).
- Amior, Michael (Oct. 2025). “The Contribution of Immigration to Local Labor Market Adjustment”. en. In: *Journal of Labor Economics* 43.4, pp. 1169–1206. ISSN: 0734-306X, 1537-5307. DOI: [10.1086/730163](https://www.journals.uchicago.edu/doi/10.1086/730163). URL: <https://www.journals.uchicago.edu/doi/10.1086/730163> (visited on 05/11/2026).
- Aubry, Amandine et al. (Feb. 2026). “Does immigration affect native wages? A meta-analysis”. en. In: *Labour Economics* 98, p. 102815. ISSN: 09275371. DOI: [10.1016/](https://doi.org/10.1016/j.labeco.2025.102815)

- j.labeco.2025.102815. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0927537125001393> (visited on 06/16/2026).
- Bartel, Ann P. (1989). “Where Do the New U.S. Immigrants Live?” In: *Journal of Labor Economics* 7.4, pp. 371–391. DOI: [10.1086/298213](https://doi.org/10.1086/298213). eprint: <https://www.journals.uchicago.edu/doi/pdf/10.1086/298213>. URL: <https://www.journals.uchicago.edu/doi/abs/10.1086/298213>.
- Basso, Gaetano and Giovanni Peri (Aug. 2020). “Internal Mobility: The Greater Responsiveness of Foreign-Born to Economic Conditions”. en. In: *Journal of Economic Perspectives* 34.3, pp. 77–98. ISSN: 0895-3309. DOI: [10.1257/jep.34.3.77](https://doi.org/10.1257/jep.34.3.77). URL: <https://pubs.aeaweb.org/doi/10.1257/jep.34.3.77> (visited on 02/13/2026).
- Beine, Michel, Simone Bertoli, and Jesús Fernández-Huertas Moraga (Apr. 2016). “A Practitioners’ Guide to Gravity Models of International Migration”. en. In: *The World Economy* 39.4, pp. 496–512. ISSN: 0378-5920, 1467-9701. DOI: [10.1111/twec.12265](https://doi.org/10.1111/twec.12265). URL: <https://onlinelibrary.wiley.com/doi/10.1111/twec.12265> (visited on 02/10/2026).
- Borjas, George J. (2001). “Does Immigration Grease the Wheels of the Labor Market?” en. In: pp. 69–119. DOI: [10.1111/j.1542-4774.2012.01052.x](https://doi.org/10.1111/j.1542-4774.2012.01052.x).
- Braun, Marius and Jarom Görts (Mar. 2025). “Differences in Labour Mobility Between Immigrant Generations: Evidence From the European Debt Crisis”. en. In: *Population, Space and Place* 31.2, e2884. ISSN: 1544-8444, 1544-8452. DOI: [10.1002/psp.2884](https://doi.org/10.1002/psp.2884). URL: <https://onlinelibrary.wiley.com/doi/10.1002/psp.2884> (visited on 11/09/2025).
- Cadena, Brian C. (Mar. 2014). “Recent immigrants as labor market arbitrageurs: Evidence from the minimum wage”. en. In: *Journal of Urban Economics* 80, pp. 1–12. ISSN: 00941190. DOI: [10.1016/j.jue.2013.10.002](https://doi.org/10.1016/j.jue.2013.10.002). URL: <https://linkinghub.elsevier.com/retrieve/pii/S0094119013000806> (visited on 02/13/2026).
- Cadena, Brian C. and Brian K. Kovak (Jan. 2016). “Immigrants Equilibrate Local Labor Markets: Evidence from the Great Recession”. en. In: *American Economic Journal: Applied Economics* 8.1, pp. 257–290. ISSN: 1945-7782, 1945-7790. DOI: [10.1257/app.20140095](https://doi.org/10.1257/app.20140095). URL: <https://pubs.aeaweb.org/doi/10.1257/app.20140095> (visited on 02/13/2026).
- Card, David (2001). “Immigrant Inflows, Native Outflows, and the Local Labor Market Impacts of Higher Immigration”. en. In: *Journal of Labor Economics*.
- Ganong, Peter and Daniel Shoag (Nov. 2017). “Why has regional income convergence in the U.S. declined?” en. In: *Journal of Urban Economics* 102, pp. 76–90. ISSN: 00941190. DOI: [10.1016/j.jue.2017.07.002](https://doi.org/10.1016/j.jue.2017.07.002). URL: <https://linkinghub.elsevier.com/retrieve/pii/S0094119017300591> (visited on 05/06/2026).

- Jaeger, David A, Joakim Ruist, and Jan Stuhler (Feb. 2018). “Shift-Share Instruments and the Impact of Immigration”. en. In: *NBER*.
- Jaeger, David A. (Sept. 2007). “Green Cards and the Location Choices of Immigrants in the United States, 1971–2000”. In: *Immigration*. Ed. by Barry R. Chiswick. Emerald Group Publishing Limited. ISBN: 978-0-76231-391-4. DOI: [10.1016/S0147-9121\(07\)00004-0](https://doi.org/10.1016/S0147-9121(07)00004-0). eprint: https://www.emerald.com/book/chapter-pdf/8875080/s0147-9121_07_00004-0.pdf. URL: [https://doi.org/10.1016/S0147-9121\(07\)00004-0](https://doi.org/10.1016/S0147-9121(07)00004-0).
- Maré, David C, Melanie Morten, and Steven Stillman (2007). “Settlement Patterns and the Geographic Mobility of Recent Migrants to New Zealand”. en. In.
- McFadden, Daniel (1974). *Conditional Logit Analysis of Qualitative Choice Behavior*. Book Section (Book Chapter). DOI: [10.1016/S0304-4076\(00\)00002-3](https://doi.org/10.1016/S0304-4076(00)00002-3).
- McKenzie, David and Hillel Rapoport (Nov. 2010). “Self-Selection Patterns in Mexico-U.S. Migration: The Role of Migration Networks”. en. In: *Review of Economics and Statistics* 92.4, pp. 811–821. ISSN: 0034-6535, 1530-9142. DOI: [10.1162/REST_a_00032](https://doi.org/10.1162/REST_a_00032). URL: <https://direct.mit.edu/rest/article/92/4/811-821/57849> (visited on 06/08/2026).
- Munshi, K. (May 2003). “Networks in the Modern Economy: Mexican Migrants in the U. S. Labor Market”. en. In: *The Quarterly Journal of Economics* 118.2, pp. 549–599. ISSN: 0033-5533, 1531-4650. DOI: [10.1162/003355303321675455](https://doi.org/10.1162/003355303321675455). URL: <https://academic.oup.com/qje/article-lookup/doi/10.1162/003355303321675455> (visited on 02/19/2026).
- Postel, Hannah M. (Feb. 2026). “Asian Immigration to the United States in Historical Perspective”. en. In: *Journal of Economic Perspectives* 40.1, pp. 191–214. ISSN: 0895-3309, 1944-7965. DOI: [10.1257/jep.20251453](https://doi.org/10.1257/jep.20251453). URL: <https://pubs.aeaweb.org/doi/10.1257/jep.20251453> (visited on 06/04/2026).
- Schündeln, Matthias (Jan. 2014). “ARE IMMIGRANTS MORE MOBILE THAN NATIVES? EVIDENCE FROM GERMANY”. en. In: *Journal of Regional Science* 54.1, pp. 70–95. ISSN: 0022-4146, 1467-9787. DOI: [10.1111/jors.12072](https://doi.org/10.1111/jors.12072). URL: <https://onlinelibrary.wiley.com/doi/10.1111/jors.12072> (visited on 03/02/2026).
- Silva, J. M. C. Santos and Silvana Tenreyro (Nov. 2006). “The Log of Gravity”. en. In: *The Review of Economics and Statistics* 88.4, pp. 641–658. ISSN: 0034-6535, 1530-9142. DOI: [10.1162/rest.88.4.641](https://doi.org/10.1162/rest.88.4.641). URL: <https://direct.mit.edu/rest/article/88/4/641/57668/The-Log-of-Gravity> (visited on 07/02/2026).
- Sjaastad, Larry A. (Oct. 1962). “The Costs and Returns of Human Migration”. en. In: *Journal of Political Economy* 70.5, Part 2, pp. 80–93. ISSN: 0022-3808, 1537-534X.

DOI: 10.1086/258726. URL: <https://www.journals.uchicago.edu/doi/10.1086/258726> (visited on 03/02/2026).

Table 10: Reduced form estimates

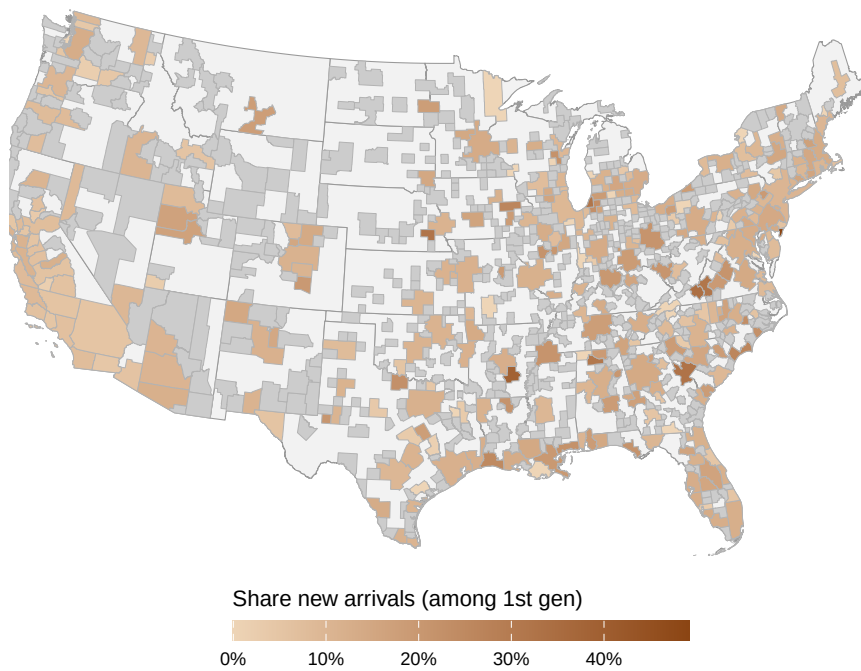
State move	1
Model:	(1)
<i>Variables</i>	
First generation migrant	0.0056*** (0.0012)
Second generation migrant	0.0018 (0.0016)
Years of education / 100	0.1602*** (0.0157)
Age / 100	-0.4462*** (0.0541)
Age squared / 100	0.0043*** (0.0006)
Male	0.0036*** (0.0006)
Mean dep. var. — natives	0.01857
<i>Fit statistics</i>	
Observations	234,203
R ²	0.00962

Clustered (cpsid) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Note: State move (1-year) in the 1995, 2005, 2015, 2020, and 2025 CPS-ASEC waves. Models include state and year fixed effects.

Figure 10: Share of newly arrived migrants over the foreign-born population.



Source: IPUMS ASEC CPS. New arrivals: ≤ 5 years since immigration.

Note: Sample restricted to first-generation migrants residing in a metropolitan area with a valid code. New arrivals are defined as those who entered the US 5 years or less before the survey. The share is computed over the total first-generation population in each MSA, aggregated across all survey years and weighted by ASEC sample weights.

Table 11: Reduced form estimates – Male sample

Model:	State move (1)	State move (5 yr) (2)	Division move (5 yr) (3)
<i>Variables</i>			
First generation migrant	0.0059*** (0.0006)	0.0142*** (0.0025)	0.0113*** (0.0023)
Second generation migrant	-0.0001 (0.0008)	-0.0003 (0.0036)	0.0001 (0.0035)
Years of education / 100	0.1997*** (0.0077)	0.8552*** (0.0347)	0.6634*** (0.0338)
Age / 100	-0.5255*** (0.0265)	-1.1150*** (0.1207)	-0.6176*** (0.1122)
Age squared / 100	0.0053*** (0.0003)	0.0098*** (0.0015)	0.0048*** (0.0014)
Mean dep. var. — natives	0.01942	0.08078	0.05142
<i>Fit statistics</i>			
Observations	871,637	132,030	107,134
R ²	0.01037	0.03304	0.02252

Clustered (cpsid) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Note: Sample restricted to male respondents. All specifications include state and year fixed effects.

Table 12: Reduced form estimates – Female sample

Model:	State move (1)	State move (5 yr) (2)	Division move (5 yr) (3)
<i>Variables</i>			
First generation migrant	0.0031*** (0.0006)	0.0060** (0.0028)	0.0068** (0.0026)
Second generation migrant	-0.0001 (0.0009)	-0.0042 (0.0039)	-0.0008 (0.0038)
Years of education / 100	0.1450*** (0.0081)	0.6599*** (0.0394)	0.4875*** (0.0382)
Age / 100	-0.5847*** (0.0284)	-1.3783*** (0.1292)	-1.0500*** (0.1239)
Age squared / 100	0.0061*** (0.0003)	0.0131*** (0.0016)	0.0101*** (0.0015)
Mean dep. var. — natives	0.01748	0.07605	0.04819
<i>Fit statistics</i>			
Observations	674,143	102,173	83,956
R ²	0.01006	0.02969	0.02168

Clustered (cpsid) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Note: Sample restricted to female respondents. All specifications include state and year fixed effects.

Table 13: Reduced form estimates – Heterogeneity by skill level

Model:	State move (1)	State move (5 yr) (2)	Division move (5 yr) (3)
<i>Variables</i>			
First generation migrant	0.0058*** (0.0008)	0.0086** (0.0035)	0.0123*** (0.0033)
Second generation migrant	0.0001 (0.0010)	-0.0018 (0.0042)	0.0024 (0.0040)
Less than high school	0.0456*** (0.0071)	0.1624*** (0.0266)	0.1341*** (0.0192)
Some high school	0.0142*** (0.0017)	0.0349*** (0.0077)	0.0392*** (0.0072)
High school graduate	0.0068*** (0.0011)	0.0178*** (0.0051)	0.0262*** (0.0049)
Some college	0.0041*** (0.0008)	0.0120*** (0.0039)	0.0172*** (0.0036)
First generation migrant × Less than high school	-0.0039 (0.0066)	-0.0081 (0.0235)	0.0144 (0.0159)
First generation migrant × Some high school	-0.0058*** (0.0016)	-0.0027 (0.0070)	-0.0038 (0.0063)
First generation migrant × High school graduate	-0.0045*** (0.0010)	-0.0053 (0.0045)	-0.0127*** (0.0042)
First generation migrant × Some college	-0.0042*** (0.0011)	-0.0125** (0.0049)	-0.0215*** (0.0044)
Second generation migrant × Less than high school	0.0048 (0.0137)	0.0559 (0.0651)	-0.0263* (0.0141)
Second generation migrant × Some high school	-0.0015 (0.0038)	-0.0025 (0.0128)	-0.0016 (0.0132)
Second generation migrant × High school graduate	-0.0010 (0.0014)	-0.0027 (0.0061)	-0.0073 (0.0058)
Second generation migrant × Some college	-0.0008 (0.0014)	-0.0020 (0.0062)	-0.0063 (0.0058)
Mean dep. var. — natives ; High school	0.01857	0.07872	0.05
<i>Fit statistics</i>			
Observations	1,508,983	228,947	186,995
R ²	0.01055	0.03288	0.02367

Clustered (cpsid) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Note: Interaction terms between migrant generation and skill level. All specifications include state and year fixed effects.

Table 14: Heterogeneity by skill level, expanded FE structure.

Model:	State move (1)	State move (5 yr) (2)	Division move (5 yr) (3)
<i>Variables</i>			
First generation migrant	0.0062*** (0.0008)	0.0101*** (0.0036)	0.0128*** (0.0034)
Second generation migrant	0.0005 (0.0010)	-0.0005 (0.0043)	0.0026 (0.0041)
First generation migrant $\times$ Less than high school	0.0019 (0.0077)	-0.0077 (0.0224)	0.0144 (0.0171)
First generation migrant $\times$ Some high school	-0.0050*** (0.0018)	-0.0031 (0.0082)	-0.0021 (0.0076)
First generation migrant $\times$ High school graduate	-0.0053*** (0.0011)	-0.0086* (0.0049)	-0.0135*** (0.0045)
First generation migrant $\times$ Some college	-0.0051*** (0.0012)	-0.0146*** (0.0052)	-0.0215*** (0.0047)
Second generation migrant $\times$ Less than high school	0.0105 (0.0140)	0.0509 (0.0556)	-0.0331 (0.0256)
Second generation migrant $\times$ Some high school	-0.0006 (0.0039)	-0.0040 (0.0137)	0.0001 (0.0140)
Second generation migrant $\times$ High school graduate	-0.0016 (0.0015)	-0.0056 (0.0064)	-0.0073 (0.0061)
Second generation migrant $\times$ Some college	-0.0017 (0.0014)	-0.0042 (0.0064)	-0.0064 (0.0061)
Mean dep. var. — natives High school	0.01857	0.07872	0.05
<i>Fit statistics</i>			
Observations	1,508,983	228,944	186,992
R ²	0.01090	0.03455	0.02528

Clustered (cpsid) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Note: Same specification as Table 13, replacing the state and year fixed effects with state $\times$ skill and year fixed effects, so that the generation $\times$ skill interaction is identified net of any state-specific skill composition. Standard errors clustered at the household level.

Table 15: Reduced form estimates – Heterogeneity by ethnicity

Model:	State move (1)	State move (5 yr) (2)	Division move (5 yr) (3)
<i>Variables</i>			
First generation migrant	-0.0020 (0.0014)	-0.0082 (0.0065)	-0.0021 (0.0061)
Second generation migrant	-0.0018 (0.0020)	-0.0086 (0.0089)	-0.0029 (0.0083)
First generation migrant $\times$ White non-Hispanic	0.0071*** (0.0017)	0.0187** (0.0078)	0.0076 (0.0074)
First generation migrant $\times$ Black non-Hispanic	0.0078*** (0.0023)	0.0326*** (0.0105)	0.0209** (0.0095)
First generation migrant $\times$ Asian non-Hispanic	0.0079*** (0.0025)	0.0139 (0.0114)	0.0137 (0.0105)
First generation migrant $\times$ Mexican Hispanic	0.0054*** (0.0017)	0.0222*** (0.0078)	0.0084 (0.0074)
Second generation migrant $\times$ White non-Hispanic	0.0026 (0.0022)	0.0188* (0.0100)	0.0121 (0.0096)
Second generation migrant $\times$ Black non-Hispanic	0.0048 (0.0037)	0.0207 (0.0162)	0.0145 (0.0143)
Second generation migrant $\times$ Asian non-Hispanic	0.0023 (0.0032)	0.0016 (0.0138)	0.0059 (0.0127)
Second generation migrant $\times$ Mexican Hispanic	0.0003 (0.0023)	-0.0016 (0.0101)	-0.0064 (0.0094)
Mean dep. var. — natives White non - hispanic	0.01857	0.07872	0.05
<i>Fit statistics</i>			
Observations	1,517,107	229,978	187,476
R ²	0.01153	0.03675	0.02848

Clustered (cpsid) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Note: Interaction terms between migrant generation and ethnicity. All specifications include state and year fixed effects.

Table 16: Gravity model estimates with log average rent as an additional control.

Dependent Variable:	flow			
Model:	Baseline	Multilateral	Heterogeneous	Corridor
	(1)	(2)	(3)	(4)
<i>Variables</i>				
Log distance	-0.5603*** (0.0490)	-0.5956*** (0.0460)	-0.6400*** (0.0503)	
Border	0.6918*** (0.0977)	0.5430*** (0.0689)	0.5102*** (0.0712)	
Foreign enclave (Bartik)	0.0620* (0.0319)	-0.0561* (0.0328)	-0.0706** (0.0354)	-0.3976 (0.2980)
Internal enclave (Bartik)	0.1508*** (0.0277)	0.2514*** (0.0235)	0.2630*** (0.0260)	-0.5212*** (0.1387)
Population / 1000	0.0002*** (0.0000)			
Share of graduates	1.7517*** (0.6150)			
Employment rate	1.8222 (1.7832)			
Log average wage	-0.5810 (0.4078)			
Log average rent	-0.0980 (0.2902)			
Share of migrants / population	0.5192 (0.6126)			
Log distance - 1st gen			0.1525** (0.0680)	
Log distance - 2nd gen			0.2496** (0.1038)	
Border - 1st gen			0.0788 (0.1056)	
Border - 2nd gen			0.1101 (0.1421)	
Foreign enclave - 1st gen			0.0812 (0.0560)	0.0590 (0.0538)
Foreign enclave - 2nd gen			0.0539 (0.0719)	0.0279 (0.0747)
Internal enclave - 1st gen			-0.0493 (0.0307)	-0.0292 (0.0196)
Internal enclave - 2nd gen			0.0135 (0.0455)	0.0088 (0.0297)
<i>Fixed-effects</i>				
origin-year-generation	Yes	Yes	Yes	Yes
destination-year-generation		Yes	Yes	Yes
origin-destination				Yes
<i>Fit statistics</i>				
Observations	136,250	114,421	114,421	93,916

Clustered (od) standard-errors in parentheses

Signif. Codes: ***, 0.01, **, 0.05, *, 0.1

Note: Dependent variable is bilateral migration flow, estimated via PPML. Same specification as Table 8; Column 1 additionally includes log average rent as a destination pull factor, to assess whether the log average wage coefficient reflects a compensating differential for local cost of living. Log average rent is a destination-level control and is therefore absorbed once destination-year-generation fixed effects are included in Columns 2–4, which are otherwise unchanged from Table 8. Standard errors clustered at the origin-destination pair level in parentheses.

Table 17: Ownership status probability.

Model:	Owner (1)
<i>Variables</i>	
First generation migrant	-0.1200*** (0.0018)
Second generation migrant	0.0136*** (0.0023)
Years of education / 100	2.1476*** (0.0212)
Age / 100	4.6909*** (0.0700)
Age squared / 100	-0.0430*** (0.0009)
Male	0.0184*** (0.0009)
Mean dep. var. — natives	0.7252
<i>Fit statistics</i>	
Observations	1,545,780
R ²	0.11654

Clustered (cpsid) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Note: Linear probability model for home ownership (1 = owns residence, 0 = rents) on migrant generation, years of education, age, age squared and sex, with state and year fixed effects. Standard errors clustered at the household level. Mean dependent variable for natives is 0.7252.